

Rate-Distortion Theory of Robust Adaptive Steganographic Coding

Qiyi Yao , Member, IEEE, Weiming Zhang , Kai Zeng , and Kejiang Chen , Senior Member, IEEE

Abstract—As an important technique to achieve covert communication, steganography has developed greatly in pursuit of the secrecy of concealment suggested by Shannon. The most widely adopted approach, adaptive steganography, consists of two phases: distortion calculation phase and adaptive steganographic coding phase, with the latter being the critical phase through which the secret message is embedded into the cover signal. Conventionally, adaptive steganography assumes a noise-free, lossless channel between the sender and receiver. However, in many practical real-world applications, the stego signal (cover signal with secret message embedded) would suffer from various lossy operations and be modified during transmission, leading to extraction errors at the receiver. To address this issue, robust adaptive steganography is essential with robust adaptive steganographic coding playing a key role. In this paper, we investigate the rate-distortion theory of robust adaptive steganographic coding. We formalize it as determining the supremum of all achievable *embedding rates* of a specialized communication system with a *normalized adaptive distortion function* and a non-stationary, memoryless sequence of discrete channels (DMCs), ensuring error-free transmission of the secret message. The *rate-distortion bound* is established, representing a significant advancement in the field of robust steganography. Furthermore, we show that the existing robust adaptive steganographic codes perform far from the bound, highlighting the need for further research and development in this area. By modeling real-world noisy channels as non-stationary DMCs, our bound sets the theoretical limit for designing robust adaptive steganographic codes that can be applied to various real-world scenarios including image, audio, and video steganography.

Index Terms—Robust adaptive steganographic coding, normalized adaptive distortion function, non-stationary memoryless sequence of discrete channels, rate-distortion bound, i.n.i.d. RVs.

I. INTRODUCTION

THERE are three general types of secrecy systems suggested by Shannon in his profound work [1]: concealment

systems, privacy systems, and “true” secrecy systems. As a branch of information hiding, steganography is in pursuit of the secrecy of concealment, so that covert communications can be conducted through the technique [2], [3], [4], [5], [6], [7]. The most widely used scheme is called adaptive steganography [8] which is composed of two phases: the distortion calculation phase and the adaptive steganographic coding phase. From the perspective of the adaptive steganographic coding phase, the distortion calculation phase generates an *adaptive distortion function* according to the characteristics of the cover signal. Then, adaptive steganographic coding embeds secret messages into the cover signal subject to the adaptive distortion function and obtains the stego signal. The stego signal is sent to the receiver, from which the secret message is extracted.

Adaptive steganography has many applications characterized by different cover media, such as image steganography [9], [10], [11], [12], [13], audio steganography [14], [15], and video steganography [16], [17], [18], [19]. All these applications employ the adaptive steganographic coding algorithm called Syndrome-Trellis Coding (STC) [20] which is designed based on linear convolutional codes and the Viterbi decoding algorithm [21]. Aside from STC, Steganographic Polar Coding (SPC) [22], [50] is another near-optimal adaptive steganographic coding method which is designed based on polar codes [23] and the Successive Cancellation List (SCL) decoding algorithm [24]. SPC performs better than STC when the cover length N is relatively large. It is also proven that polar codes are optimal in adaptive steganography under the constant distortion profile (Hamming distortion measure) given the exact Bhattacharyya parameters of the coordinate channels [25]. Recently, another adaptive steganographic coding method based on Low-Density Generator-Matrix (LDGM) codes [26] has been developed which also shows near-optimal performance and has its advantage in the large payload scenario [27].

Conventionally, adaptive steganography assumes a noise-free lossless channel between the sender and receiver. However, in real-world applications, the stego (cover signal with secret message embedded) would suffer from various lossy operations and be modified during transmission, leading to extraction error at the receiver. To make steganography practical, numerous robust adaptive steganography methods have been developed, which can be categorized into two approaches: robust cover construction [28], [29] and cover preprocessing [30], [31]. At the sender, these methods first exploit the intrinsic properties of specific noisy channels or lossy operations, such as JPEG compression, to adjust the original cover for noise mitigation [32]. Then,

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Qiyi Yao was with the CAS Key Laboratory of Electro-magnetic Space Information, Anhui Province Key Laboratory of Digital Security, University of Science and Technology of China, Hefei 230026, China. He is now with the Department of Electrical and Electronics Engineering, Bilkent University, Ankara 06800, Türkiye (e-mail: qyyao@mail.ustc.edu.cn).

Weiming Zhang and Kejiang Chen are with the CAS Key Laboratory of Electro-magnetic Space Information, Anhui Province Key Laboratory of Digital Security, University of Science and Technology of China, Hefei 230026, China (e-mail: zhangwm@ustc.edu.cn; chenkj@ustc.edu.cn).

Kai Zeng is with the Department of Information Engineering and Mathematics, University of Siena, 53100 Siena, Italy (e-mail: kai.zeng@unisi.it).

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robust adaptive steganographic coding is employed for all robust steganography methods to counteract noise introduced during transmission while embedding the secret message. This paper focuses on robust adaptive steganographic coding, as it plays a critical role in ensuring the reliability of robust steganography under such conditions.

Technically, robust adaptive steganographic coding methods can be divided into two classes distinguished by where error correction is conducted. In the first class, the error correction is conducted on the secret message [28], [29], [30]. Then, the secret message with error correction is embedded into the cover signal using a conventional adaptive steganographic coding algorithm. When doing the extraction, the receiver first extracts the secret message with error correction using the conventional procedure and then decodes the extracted message to obtain the “true” secret message. In the second class, the error correction is conducted on the stego [33], [34], [35], [36]. Note that if error correction is directly conducted on the stego, the length of the stego would increase, making it obviously different from the cover, in which case the purpose of concealment is breached. Therefore, these methods are usually based on elaborately designed nested codes [37]. When doing the extraction, the receiver first decodes the received sequence and obtains the “true” stego. Then, by conducting the conventional adaptive steganographic extraction, the secret message is obtained.

All these existing robust adaptive steganographic coding methods focus on developing techniques to embed secret messages while achieving high robustness and low detection rates. However, they are evaluated experimentally against one another and lack a clear theoretical limit to guide the design of such algorithms. In this paper, we aim to provide a theoretical basis for robust adaptive steganographic coding. We formalize it as determining the supremum of all achievable *embedding rates* of a specialized communication system characterized by a *normalized adaptive distortion function* and a non-stationary, memoryless sequence of discrete channels, ensuring error-free transmission of the secret message. We establish the theoretical *rate-distortion bound* of the problem and give an example of the traditional case where each component in the system has a single-letter characterization to illustrate that our result is compatible with the classical bound of information hiding [38]. By modeling noisy channels in real-world applications into non-stationary discrete memoryless channels (DMCs), our bound provides a benchmark for evaluating existing robust methods and sets the theoretical limit for designing new practical robust adaptive steganographic coding schemes. Specifically, we present experimental evaluations of the existing robust adaptive steganographic coding methods against the established rate-distortion bound for the binary Hamming case which is widely used in the literature [20], [22], [27], [33], [36]. This setting is practical and relevant for applications such as steganography for backdoor attacks in machine learning [39], Internet of Things (IoT) [40], [41], and stegomalware [42]. The experimental results show that the existing methods perform far from the bound, highlighting the need for further research and development in this area.

The remainder of the paper is organized as follows. In the next section, the basic notation along with some fundamental

concepts and properties is introduced. In Section III, we formalize the problem of robust adaptive steganographic coding. In Section IV, we present the main result of the paper regarding the rate-distortion bound of the problem of robust adaptive steganographic coding and provide an example of the traditional single-letter case. Moreover, we also show that the existing robust adaptive steganographic coding methods perform far from the rate-distortion bound experimentally in this section. The proof of the main theorem is provided in Section V. Section VI contains some concluding remarks along with some future directions.

II. PRELIMINARIES

We use capital letters to denote random variables (e.g., X). The corresponding lowercase letters (e.g., x) and script letters (e.g., $\mathcal{X}$) denote the individual values and the alphabets (or ranges) of the random variables (RVs), respectively. We assume that all RVs take on values from finite alphabets in this paper except when otherwise indicated. The notation $|\mathcal{X}|$ is used to denote the cardinality of set $\mathcal{X}$. Notations $\mathbb{E}[X]$ and $\text{Var}[X]$ denote the expectation and variance of X , respectively. Random sequences of countably infinite length are denoted by bold-face capital letters (e.g., $\mathbf{X} = (X_1, X_2, \dots)$) while their outcomes are denoted by bold-face lowercase letters (e.g., $\mathbf{x} = (x_1, x_2, \dots)$). We assume that the cardinality sequences (e.g., $\{|\mathcal{X}_t|\}_{t=1}^{\infty}$) of the alphabets of the random sequences are bounded in this paper. For any $i, j \in \mathbb{N}^+, i \leq j$, we write X_i^j to denote the partial random vector $(X_i, \dots, X_j)$. Correspondingly, $x_i^j = (x_i, \dots, x_j)$. The probability assignment of an RV X is denoted by P_X . The joint probability assignment of RVs (X, Y) is denoted by $P_{X,Y}$. Similarly, the conditional probability assignment is written in the form $P_{Y|X}$. We use the standard notation $H(\cdot)$ and $I(\cdot; \cdot)$ as well as $H(\cdot|\cdot)$ and $I(\cdot; \cdot|\cdot)$ to denote Shannon's entropy, mutual information, and their conditional form, respectively.

We focus on the general non-stationary memoryless scenario where random sequences and random vectors considered in this paper are independent but not necessarily identically distributed (i.n.i.d.). Thereby, for an i.n.i.d. random sequence $\mathbf{X}$, we define its normalized average entropy as

$$\overline{H(\mathbf{X})} \triangleq \lim_{N \rightarrow \infty} \frac{1}{N} H(X_1^N) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N H(X_t), \quad (1)$$

where we assume that the limits exist.¹

Correspondingly, given i.n.i.d. random sequences $\mathbf{X}$ and $\mathbf{Y}$ such that $P_{\mathbf{X}, \mathbf{Y}} = \otimes_t P_{X_t, Y_t}$, i.e., the sequence $(\mathbf{X}, \mathbf{Y}) = \{(X_t, Y_t)\}_{t=1}^{\infty}$ is i.n.i.d., the normalized average mutual information between $\mathbf{X}$ and $\mathbf{Y}$ is defined as

$$\overline{I(\mathbf{X}; \mathbf{Y})} \triangleq \lim_{N \rightarrow \infty} \frac{1}{N} I(X_1^N; Y_1^N) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N I(X_t; Y_t), \quad (2)$$

where we also assume that the limits exist.

¹The normalized average entropy $\overline{H(\mathbf{X})}$ is also known as the *entropy rate* of the stochastic process $\mathbf{X} = \{X_t\}_t$.

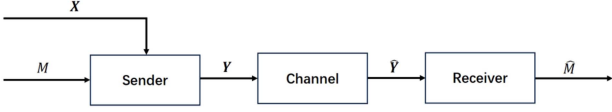


Fig. 1. The communication system modeling the problem of robust adaptive steganographic coding. The signals M , X , Y , $\hat{Y}$, $\hat{M}$ are respectively, the secret message signal, the cover signal, the stego signal, the channel output, and the extracted secret message.

In the last of this section, we present Chebyshev's inequality [43], [44] as a lemma which will be useful in the later sections when we prove our main results.

Lemma 1 (Chebyshev's Inequality): Given real-valued random variable X that has finite non-zero variance σ^2 , $\forall k \in \mathbb{R}^+$,

$$\Pr(|X - \mathbb{E}[X]| \geq k\sigma) \leq \frac{1}{k^2}. \quad (3)$$

III. PROBLEM FORMULATION

Let $\mathbf{X} = \{X_t\}_t$ be the i.i.d. cover random sequence with finite alphabets $\{\mathcal{X}_t\}_t$ and probability distributions $\{P_{X_t}\}_t$. Let M be the secret message signal that is independent of the cover signal $\mathbf{X}$. The robust adaptive steganographic coding problem of interest is modeled as a communication system depicted in Fig. 1. The sender produces the stego signal $\mathbf{Y}$ by embedding the secret message M into the cover signal $\mathbf{X}$ subject to a normalized adaptive distortion function $\overline{D}(\cdot, \cdot) : (\mathcal{X}_1 \times \mathcal{X}_2 \times \dots) \times (\mathcal{Y}_1 \times \mathcal{Y}_2 \times \dots) \rightarrow [0, \infty)$ defined as

$$\overline{D}(\mathbf{x}, \mathbf{y}) \triangleq \lim_{N \rightarrow \infty} \frac{1}{N} D_1^N(x_1^N, y_1^N) \triangleq \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N D_t(x_t, y_t), \quad (4)$$

that is applied to measure the distortion between the cover signal $\mathbf{X} = (X_1, X_2, \dots)$ and the stego signal $\mathbf{Y} = (Y_1, Y_2, \dots)$ where $\mathbf{x} = (x_1, x_2, \dots)$, $\mathbf{y} = (y_1, y_2, \dots)$, and $\forall t \in \mathbb{N}^+$, $x_t \in \mathcal{X}_t$, $y_t \in \mathcal{Y}_t$. Each $D_t(\cdot, \cdot) : \mathcal{X}_t \times \mathcal{Y}_t \rightarrow [0, \infty)$ in (4) is a bounded single-letter distortion measure and the existence of the limits is assumed. Indeed, an adaptive distortion function is a sequence of single-letter distortion measures.

Definition 1: A normalized adaptive distortion function $\overline{D}(\cdot, \cdot) : \mathcal{X} \times \mathcal{Y} \rightarrow [0, \infty)$ is called *uniformly bounded* if there exists $\gamma_{\max} \in \mathbb{R}^+$ such that $\forall t \in \mathbb{N}^+$, $\forall x_t \in \mathcal{X}_t$, $y_t \in \mathcal{Y}_t$, $D_t(x_t, y_t) \leq \gamma_{\max}$, where $\mathcal{X} = \prod_{t=1}^{\infty} \mathcal{X}_t$, $\mathcal{Y} = \prod_{t=1}^{\infty} \mathcal{Y}_t$. $\gamma_{\max}$ is called the uniform bound of $\overline{D}(\cdot, \cdot)$.

All the normalized adaptive distortion functions considered in this paper are assumed to be uniformly bounded except when otherwise indicated. Besides, the cardinality sequences $\{|\mathcal{X}_t|\}_{t=1}^{\infty}$ and $\{|\mathcal{Y}_t|\}_{t=1}^{\infty}$ are both assumed to be bounded which fits the real-world scenarios. For instance, the cardinality sequences are bounded by 256 for grayscale images.

After message embedding, the stego signal $\mathbf{Y}$ passes through the probabilistic noisy channel and becomes $\hat{\mathbf{Y}}$ where the cardinality sequence $\{|\hat{\mathcal{Y}}_t|\}_{t=1}^{\infty}$ is bounded. Then, the receiver extracts the reconstructed secret message $\hat{M}$ from $\hat{\mathbf{Y}}$. As we

are considering the general non-stationary memoryless scenario, a non-stationary memoryless sequence of discrete channels $\{W_t\}_t$ with transition probabilities $\{\{P_{\hat{Y}_t|Y_t}(\hat{y}_t|y_t) : y_t \in \mathcal{Y}_t, \hat{y}_t \in \hat{\mathcal{Y}}_t\}\}_t$ is considered. Compared to the traditional setting where the cover is i.i.d. and the channel is stationary and memoryless, our setting is much more general with the traditional setting being a special case. We make a comment here that the concept of adaptive steganography [20] came up much later than the proposition of the traditional information embedding theorem, and the most important component of adaptive steganography is the normalized adaptive distortion function $\overline{D}(\cdot, \cdot)$ which is more practical in real-world applications [14], [16], [18], [20] compared to the traditional single-letter distortion measure employed in the traditional setting [38].

The embedding rate of the system is given by $R = \frac{H(M)}{N}$. When considering an infinite cover length, the embedding rate $R = \lim_{N \rightarrow \infty} \frac{H(M)}{N}$. The rate-distortion bound of robust adaptive steganographic coding, denoted by $R_D(d)$, is defined as the supremum of all achievable embedding rates for communicating a secret message M such that the extraction error probability $\Pr(\hat{M} \neq M)$ is arbitrarily close to 0 and the average normalized distortion between cover signal $\mathbf{X}$ and stego signal $\mathbf{Y}$ is less than or equal to design distortion d when the cover length N tends to infinity.

IV. MAIN RESULT

In this section, we will present the main result regarding the rate-distortion bound of robust adaptive steganographic coding. We will also give an example of the traditional single-letter case to further illustrate that our result is compatible with the classical bound of information hiding [38]. Experimental evaluations of the existing robust adaptive steganographic coding methods against the established rate-distortion bound along with the implications of the main result will be provided at the end of this section.

A. The Rate-Distortion Bound

Given the i.i.d. cover random sequence $\mathbf{X}$, let $\mathbf{U} = \{U_1, U_2, \dots\}$ be an i.i.d. auxiliary random sequence where for each $t \in \mathbb{N}^+$, the alphabet $\mathcal{U}_t$ of U_t is an arbitrary finite set, the cardinality sequence $\{|\mathcal{U}_t|\}_{t=1}^{\infty}$ is bounded, and $(\mathbf{U}, \mathbf{X})$ is i.i.d., i.e., $P_{\mathbf{U}, \mathbf{X}} = \otimes_t P_{U_t, X_t}$. Then, for any design distortion $d \geq 0$, define set

$$\mathcal{P}_d \triangleq \left\{ \{P_{U_t|X_t}\}_{t=1}^{\infty}, \{f_t\}_{t=1}^{\infty} : \forall t \in \mathbb{N}^+, f_t : \mathcal{U}_t \times \mathcal{X}_t \rightarrow \mathcal{Y}_t, \right. \\ \left. Y_t = f_t(U_t, X_t); \mathbb{E}[\overline{D}(\mathbf{X}, \mathbf{Y})] \leq d \right\}, \quad (5)$$

where $\{P_{U_t|X_t}\}_{t=1}^{\infty}$ contain all possible conditional probability distributions, $\{f_t\}_{t=1}^{\infty}$ contain all functions mapping from $\mathcal{U}_t \times \mathcal{X}_t$ to $\mathcal{Y}_t$ for any $t \in \mathbb{N}^+$, $P_{\mathbf{U}|\mathbf{X}} = \otimes_t P_{U_t|X_t}$, and $\mathbb{E}[\overline{D}(\mathbf{X}, \mathbf{Y})]$ is defined as

$$\mathbb{E}[\overline{D}(\mathbf{X}, \mathbf{Y})] \triangleq \int_{\mathcal{X} \times \mathcal{Y}} \overline{D} dP_{\mathbf{X}, \mathbf{Y}}. \quad (6)$$

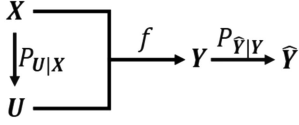


Fig. 2. The relationship of the random sequences $U, X, Y, \hat{Y}$.

For simplicity, we will also write

$$Y = f(U, X) = (f_1(U_1, X_1), f_2(U_2, X_2), \dots)$$

in the rest of the paper.

The relationship of the random sequences $U, X, Y, \hat{Y}$ is depicted in Fig. 2. The conditional distributions $\{P_{U_t|X_t}\}_{t=1}^\infty$ can be thought of as the transition probabilities of a non-stationary memoryless sequence of discrete *test channels*. As the sequence of non-stationary channels between Y and $\hat{Y}$ is also memoryless and for any fixed $f = \{f_t\}_t$, the functions $\{f_t\}_t$ do not depend on each other, the random sequence $(U, X, Y, \hat{Y}) = \{(U_t, X_t, Y_t, \hat{Y}_t)\}_t$ is i.n.i.d.

Now, we can give the main result of this paper regarding the rate-distortion bound.

Theorem 1 (Robust Adaptive Steganographic Coding): Let $X = \{X_t\}_{t=1}^\infty$ be an i.n.i.d. cover random sequence with finite alphabets $\{\mathcal{X}_t\}_{t=1}^\infty$, bounded cardinality sequence $\{|\mathcal{X}_t|\}_{t=1}^\infty$, and probability distributions $\{P_{X_t}\}_{t=1}^\infty$. Given a uniformly bounded normalized adaptive distortion function $\overline{D}(\cdot, \cdot) : \mathcal{X} \times \mathcal{Y} \rightarrow [0, \infty)$ and a non-stationary memoryless sequence of discrete channels $\{W_t\}_t$ with transition probabilities $\{\{P_{\hat{Y}_t|Y_t}(\hat{y}_t|y_t) : y_t \in \mathcal{Y}_t, \hat{y}_t \in \hat{\mathcal{Y}}_t\}\}_t$ such that cardinality sequences $\{|\mathcal{X}_t|\}_{t=1}^\infty, \{|\mathcal{Y}_t|\}_{t=1}^\infty, \{|\hat{\mathcal{Y}}_t|\}_{t=1}^\infty$ are bounded, for any design distortion $d \geq 0$,

$$R_D(d) = \sup_{(\{P_{U_t|X_t}\}_{t=1}^\infty, \{f_t\}_{t=1}^\infty) \in \mathcal{P}_d} \overline{I}(\hat{Y}; U) - \overline{I}(X; U), \quad (7)$$

where U is an i.n.i.d. auxiliary random sequence.

B. The Traditional Single-Letter Case

Traditionally, all components in the robust steganographic coding model have a single-letter characterization. We use $X, Y, \hat{Y}, U$ to denote the cover RV, stego RV, channel output RV, and the auxiliary RV, respectively. The stationary discrete channel has transition probabilities $\{P_{\hat{Y}|Y}(\hat{y}|y)\}$. The normalized adaptive distortion function is a single-letter distortion measure. By seeing the random sequences $X, Y, \hat{Y}, U$ in our model as sequences of i.i.d. RVs $X, Y, \hat{Y}, U$, changing the non-stationary discrete memoryless channel into a stationary discrete memoryless channel, and assuming a bounded single-letter distortion measure D , the traditional scenario can be recognized as a special case in our model.

The proof of Theorem 1 will be provided in Section V. For now, let us just assume that Theorem 1 holds. Then, let $X = (X, X, \dots), Y = (Y, Y, \dots), \hat{Y} = (\hat{Y}, \hat{Y}, \dots), U = (U, U, \dots)$, where $\mathcal{X}, \mathcal{Y}, \hat{\mathcal{Y}}, \mathcal{U}$ are finite alphabets. Let $P_{\hat{Y}|Y} = \otimes P_{\hat{Y}|Y}$ and the bounded single-letter distortion measure be D ,

where

$$\overline{D}(\mathbf{x}, \mathbf{y}) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N D(x_t, y_t).$$

Then, by Theorem 1, we have

$$\begin{aligned} R_D(d) &= \sup_{(\{P_{U_t|X_t}\}_{t=1}^\infty, \{f_t\}_{t=1}^\infty) \in \mathcal{P}_d} \overline{I}(\hat{Y}; U) - \overline{I}(X; U) \\ &= \sup_{(P_{U|X}, f) \in \mathcal{P}_d^{(1)}} I(\hat{Y}; U) - I(X; U), \end{aligned} \quad (8)$$

where

$$\begin{aligned} \mathcal{P}_d^{(1)} &\triangleq \left\{ P_{U|X}, f \mid f : \mathcal{U} \times \mathcal{X} \rightarrow \mathcal{Y}, \right. \\ &\quad \left. Y = f(U, X); \mathbb{E}[D(X, Y)] \leq d \right\}. \end{aligned} \quad (9)$$

This is the same as the result in [38, Claim 1]. So, by seeing our result as an extension of the traditional single-letter scenario, our bound agrees with the traditional bound when the model is simplified to the traditional single-letter case.

C. Evaluation on Existing Methods

In this subsection, we evaluate the performance of the existing robust adaptive steganographic methods against our theoretical bound. The methods considered are the widely used STC combined with Reed-Solomon codes (STC+RS) in the first class [28], [29], [30], Dual-STC (DSTC) in the second class [33], [35], and RSPC in the second class [36].

Binary embedding is considered in the evaluation where the cover is assumed to be i.i.d. binary sequences with each bit being uniformly distributed over $\{0, 1\}$. This setting is widely used in the literature of steganography [20], [22], [27], [33], [36] where only the least significant bits (LSBs) of the cover elements can be modified. Binary embedding is also practical and relevant for applications such as steganography for backdoor attacks in machine learning [39] and stegomalware [42]. The communication channel used in the evaluation is the stationary binary symmetric channel (BSC) with crossover probability q , i.e., $\text{BSC}(q)$. The BSC models several real-world noisy channels such as the salt & pepper noise considered in image steganography for IoT [41]. Since the design of all the existing methods follows the traditional assumption that codes with high performance under the constant distortion profile also perform well under other distortion profiles [20], we mainly consider the constant distortion function (Hamming distortion measure) in the evaluation where

$$\forall 1 \leq t \leq N, \quad D_t(x_t, y_t) = \begin{cases} 0, & \text{if } x_t = y_t, \\ 1, & \text{if } x_t \neq y_t. \end{cases} \quad (10)$$

The constant distortion function is widely employed in applications such as steganography for IoT [40] where the capabilities of the embedding devices are limited and the high computational complexity for calculating adaptive distortion functions cannot be afforded.

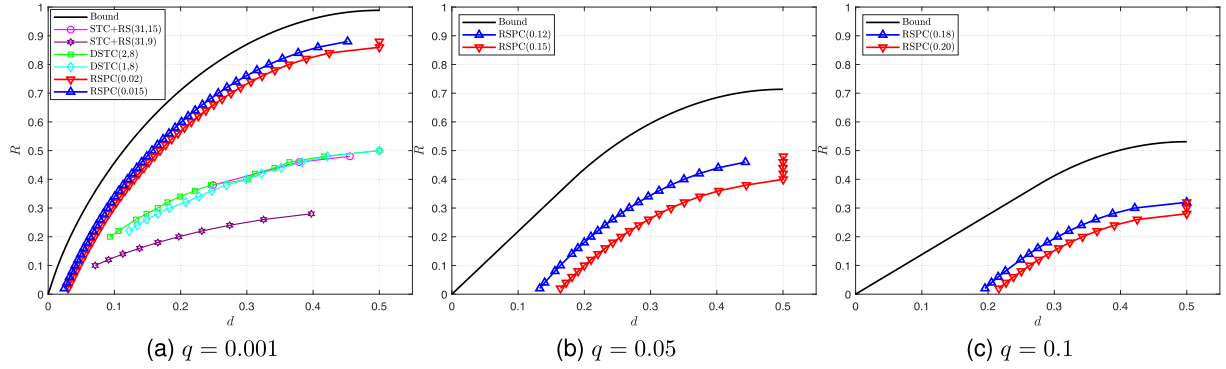


Fig. 3. Evaluation of the existing robust adaptive steganographic coding methods compared to the established rate-distortion bound for various BSCs with crossover probabilities $q = 0.001$, $q = 0.05$, and $q = 0.1$, where cover length $N = 2^{16}$.

Since our bound is compatible with the traditional single-letter bound [38], the rate-distortion bound under this binary Hamming circumstance is given by

$$R_D(d) = \begin{cases} \frac{H(d_q) - H(q)}{d_q} d, & \text{if } 0 \leq d \leq d_q, \\ H(d) - H(q), & \text{if } d_q < d \leq 0.5, \end{cases} \quad (11)$$

where $d_q = 1 - 2^{-H(q)}$. Here we note that

$$d_q \geq q, \quad \forall q \in [0, 0.5],$$

following the fact that for any $q \in [0, 0.5]$,

$$\begin{aligned} & H(q) + \log_2(1 - q) \\ &= -q \log_2 q - (1 - q) \log_2(1 - q) + \log_2(1 - q) \\ &= -q \log_2 q + q \log_2(1 - q) = q \log_2 \frac{1 - q}{q} \geq 0 \\ &\Rightarrow H(q) \geq -\log_2(1 - q) \Rightarrow 1 - 2^{-H(q)} \geq q. \end{aligned}$$

The parameters of the evaluated methods are set as follows. For STC+RS, the STC sub-matrix height is set to be 10 and the RS code with code length n^* and information length k^* is denoted as RS(n^*, k^*). DSTC with message embedding sub-matrix height h_m and error correction sub-matrix height h_s is denoted as DSTC(h_m, h_s). RSPC with robust parameter q' is denoted as RSPC(q'). All the simulation results are obtained by averaging over 100 trials and only (R, d) pairs with at least 99% extraction successful rate are counted as achievable.

Fig. 3 shows the rate-distortion performance of the evaluated methods compared to the established rate-distortion bound for different BSC crossover probabilities where the cover length $N = 2^{16}$. It can be seen from Fig. 3(a) that when the channel is relatively good with crossover probability $q = 0.001$, only RSPC performs closely to the established bound while other methods perform far from the bound. When the channel becomes worse with crossover probabilities $q = 0.05$ and $q = 0.1$, STC+RS and DSTC are not able to provide any robustness guarantee and the extraction successful rates are consistently 0%. Therefore, we do not plot their performance in Fig. 3(b) and 3(c). For RSPC, we can observe that although it can still provide robustness guarantee for channels with stronger noise, its performance degrades significantly compared to the

established bound. This indicates that there is still a large room for improvement in the design of robust adaptive steganographic codes and coding algorithms.

D. Implications of the Bound

Theorem 1 establishes a fundamental limit on the achievable rate for any robust adaptive steganographic code in the considered i.n.i.d. setting. Compared to the conventional adaptive steganographic coding scenario where the trade-off is between the embedding rate and the distortion incurred by the embedding process under the noise-free assumption, our result characterizes the trade-off between the embedding rate and the distortion incurred by the embedding process under the presence of channel noise. This is more practical and relevant for real-world applications where the channel noise is inevitable. Following (7), as the channel noise becomes stronger, the term $I(\hat{Y}; U)$ decreases and the term $I(X; U)$ remains unchanged, leading to a smaller achievable embedding rate. This is consistent with the intuition that stronger channel noise leads to worse performance of the robust adaptive steganographic coding system.

Beyond its theoretical interest, this bound serves two important practical purposes: it acts as an absolute benchmark for performance assessment, and it directly informs the structural principles that lead to optimal code constructions.

1) *Benchmarking Existing and Future Schemes:* Because the bound applies to any admissible code, it provides an unambiguous yardstick against which all existing and forthcoming methods can be measured. In our numerical evaluation (Section IV-C), we compared several representative codes from the literature with the derived rate-distortion bound. Most constructions exhibit a noticeable gap to the bound, indicating room for improvement. One notable exception is the nested coding scheme RSPC, which tracks the bound significantly more closely across the whole parameter range. This observation immediately flags nested architectures as a promising design direction worth pursuing further.

2) *From the Bound to a Constructive Design Principle:* The guidance offered by Theorem 1 becomes explicit when examining its proof. In Section V-B we construct a random binning and prove that it achieves the bound asymptotically. The nested

structure of the binning scheme indicates that optimum robust adaptive steganographic coding can potentially be achieved by nested code constructions. Thus, the bound does not merely state that a limit exists — it reveals how the limit can be reached. The key insight is that a carefully designed random binning scheme can asymptotically achieve the bound. For a practitioner, this translates into a concrete design recipe: adopt a nested code architecture and select the component codes to align with the boundary identified in Theorem 1.

V. THE PROOF

In this section, we will provide the proof of Theorem 1 which consists of two parts: the proof of achievability and the proof of the converse theorem. We will first make some preparations needed to deal with the adaptive distortion function and the non-stationary memoryless channels, and introduce several necessary tools before proving the theorem.

A. Preparation

To prove Theorem 1, we need to develop several tools which are extended from the classical information-theoretic concepts. In this subsection, we first analyze the average normalized distortion $\mathbb{E}[\overline{D}(\mathbf{X}, \mathbf{Y})]$ as defined in (6). Then, we develop the necessary tools needed for the proof.

1) *The Average Normalized Distortion:* Let $\mathbf{X} = \{X_t\}_{t=1}^\infty$, $\mathbf{Y} = \{Y_t\}_{t=1}^\infty$ be two sequences of i.i.d. RVs that take on values from finite alphabets $\{\mathcal{X}_t\}_{t=1}^\infty, \{\mathcal{Y}_t\}_{t=1}^\infty$ with bounded cardinality sequences $\{|\mathcal{X}_t|\}_{t=1}^\infty, \{|\mathcal{Y}_t|\}_{t=1}^\infty$ and probability distributions $\{P_{X_t}\}_{t=1}^\infty, \{P_{Y_t}\}_{t=1}^\infty$, respectively. Let the sequence of RV pairs $(\mathbf{X}, \mathbf{Y}) = \{(X_t, Y_t)\}_{t=1}^\infty$ be i.i.d., i.e., $P_{\mathbf{X}, \mathbf{Y}} = \otimes_t P_{X_t, Y_t}$. Given uniformly bounded normalized adaptive distortion function $\overline{D}(\cdot, \cdot) : \mathcal{X} \times \mathcal{Y} \rightarrow [0, \infty)$, for any $N \in \mathbb{N}^+$, the average normalized distortion is defined as

$$\begin{aligned} \mathbb{E}[\overline{D}_1^N(X_1^N, Y_1^N)] &\triangleq \mathbb{E}\left[\frac{1}{N} D_1^N(X_1^N, Y_1^N)\right] \\ &\triangleq \int_{\mathcal{X}_1^N \times \mathcal{Y}_1^N} \overline{D}_1^N dP_{X_1^N, Y_1^N} \\ &= \frac{1}{N} \sum_{x_1^N \in \mathcal{X}_1^N, y_1^N \in \mathcal{Y}_1^N} P_{X_1^N, Y_1^N}(x_1^N, y_1^N) D_1^N(x_1^N, y_1^N). \end{aligned} \quad (12)$$

The following proposition shows that the average normalized distortion can be decomposed.

Proposition 1: Given a normalized adaptive distortion function $\overline{D}(\cdot, \cdot)$, for any $N \in \mathbb{N}^+$,

$$\begin{aligned} \mathbb{E}[\overline{D}_1^N(X_1^N, Y_1^N)] &= \frac{1}{N} \sum_{t=1}^N \sum_{x_t \in \mathcal{X}_t, y_t \in \mathcal{Y}_t} P_{X_t, Y_t}(x_t, y_t) D_t(x_t, y_t) \\ &= \frac{1}{N} \sum_{t=1}^N \mathbb{E}[D_t(X_t, Y_t)]. \end{aligned} \quad (13)$$

Proof: Substituting (4) into (12), we have that

$$\begin{aligned} \mathbb{E}[\overline{D}_1^N(X_1^N, Y_1^N)] &= \frac{1}{N} \sum_{x_1^N \in \mathcal{X}_1^N, y_1^N \in \mathcal{Y}_1^N} P_{X_1^N, Y_1^N}(x_1^N, y_1^N) D_1^N(x_1^N, y_1^N) \\ &= \frac{1}{N} \sum_{x_1^N \in \mathcal{X}_1^N, y_1^N \in \mathcal{Y}_1^N} P_{X_1^N, Y_1^N}(x_1^N, y_1^N) \left(\sum_{t=1}^N D_t(x_t, y_t) \right) \\ &= \frac{1}{N} \sum_{x_1^N \in \mathcal{X}_1^N, y_1^N \in \mathcal{Y}_1^N} \sum_{t=1}^N P_{X_1^N, Y_1^N}(x_1^N, y_1^N) D_t(x_t, y_t) \\ &= \frac{1}{N} \sum_{t=1}^N \sum_{x_1^N \in \mathcal{X}_1^N, y_1^N \in \mathcal{Y}_1^N} P_{X_1^N, Y_1^N}(x_1^N, y_1^N) D_t(x_t, y_t) \\ &= \frac{1}{N} \sum_{t=1}^N \sum_{x_t \in \mathcal{X}_t, y_t \in \mathcal{Y}_t} D_t(x_t, y_t) \\ &\quad \cdot \sum_{x_1^{t-1}, x_{t+1}^N, y_1^{t-1}, y_{t+1}^N} P_{X_1^N, Y_1^N}(x_1^N, y_1^N) \\ &= \frac{1}{N} \sum_{t=1}^N \sum_{x_t \in \mathcal{X}_t, y_t \in \mathcal{Y}_t} P_{X_t, Y_t}(x_t, y_t) D_t(x_t, y_t), \end{aligned} \quad (14)$$

where the fourth equality follows from the special case of Tonelli's Theorem [45]. $\square$

Indeed, this result holds even when the random sequence $(\mathbf{X}, \mathbf{Y})$ is more general (e.g., a random sequence with memory) as the proof does not need anything from the joint distribution $P_{X_1^N, Y_1^N}$.

We have the following result regarding the average normalized distortion $\mathbb{E}[\overline{D}(\mathbf{X}, \mathbf{Y})]$.

Proposition 2: Given a uniformly bounded normalized adaptive distortion function $\overline{D}(\cdot, \cdot)$, the average normalized distortion $\mathbb{E}[\overline{D}(\mathbf{X}, \mathbf{Y})]$ can also be decomposed:

$$\begin{aligned} \mathbb{E}[\overline{D}(\mathbf{X}, \mathbf{Y})] &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \sum_{x_t \in \mathcal{X}_t, y_t \in \mathcal{Y}_t} P_{X_t, Y_t}(x_t, y_t) D_t(x_t, y_t) \\ &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \mathbb{E}[D_t(X_t, Y_t)]. \end{aligned} \quad (15)$$

Proof: For any $N \in \mathbb{N}^+$, we extend the domain of the function $D_1^N : \mathcal{X}_1^N \times \mathcal{Y}_1^N \rightarrow [0, \infty)$ from $\mathcal{X}_1^N \times \mathcal{Y}_1^N$ to $\mathcal{X} \times \mathcal{Y}$ where for any $(\mathbf{x}, \mathbf{y}) \in \mathcal{X} \times \mathcal{Y}$, $D_1^N(\mathbf{x}, \mathbf{y}) = D_1^N(x_1^N, y_1^N)$.

By substituting (4) into (6), we write

$$\begin{aligned} \mathbb{E}[\overline{D}(\mathbf{X}, \mathbf{Y})] &= \int_{\mathcal{X} \times \mathcal{Y}} \overline{D} dP_{\mathbf{X}, \mathbf{Y}} \\ &= \int_{\mathcal{X} \times \mathcal{Y}} \lim_{N \rightarrow \infty} \frac{1}{N} D_1^N dP_{\mathbf{X}, \mathbf{Y}}. \end{aligned} \quad (16)$$

Since $\overline{D(\cdot, \cdot)}$ is uniformly bounded, let $\gamma_{\max}$ be the uniform bound. Then, for any $N \in \mathbb{N}^+$, $\forall (\mathbf{x}, \mathbf{y}) \in \mathcal{X} \times \mathcal{Y}$,

$$\begin{aligned}\overline{D_1^N}(\mathbf{x}, \mathbf{y}) &= \frac{1}{N} D_1^N(\mathbf{x}, \mathbf{y}) = \frac{1}{N} D_1^N(x_1^N, y_1^N) \\ &= \frac{1}{N} \sum_{t=1}^N D_t(x_t, y_t) \leq \gamma_{\max}.\end{aligned}$$

We also have that

$$\int_{\mathcal{X} \times \mathcal{Y}} \gamma_{\max} dP_{\mathbf{X}, \mathbf{Y}} = \gamma_{\max},$$

which suggests that $\gamma_{\max}$ is integrable. Therefore, for any $N \in \mathbb{N}^+$, $\overline{D_1^N}$ is dominated by an integrable function $\gamma_{\max}$. By the Dominated Convergence Theorem, we can rewrite (16) as

$$\begin{aligned}\mathbb{E}[\overline{D(\mathbf{X}, \mathbf{Y})}] &= \int_{\mathcal{X} \times \mathcal{Y}} \lim_{N \rightarrow \infty} \frac{1}{N} D_1^N dP_{\mathbf{X}, \mathbf{Y}} \\ &= \lim_{N \rightarrow \infty} \int_{\mathcal{X} \times \mathcal{Y}} \frac{1}{N} D_1^N dP_{\mathbf{X}, \mathbf{Y}} \\ &= \lim_{N \rightarrow \infty} \int_{\mathcal{X}_1^N \times \mathcal{Y}_1^N} \overline{D_1^N} dP_{X_1^N, Y_1^N} \\ &= \lim_{N \rightarrow \infty} \mathbb{E}[\overline{D_1^N}(X_1^N, Y_1^N)] \\ &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \sum_{x_t \in \mathcal{X}_t, y_t \in \mathcal{Y}_t} P_{X_t, Y_t}(x_t, y_t) D_t(x_t, y_t), \quad (17)\end{aligned}$$

where the last equality follows from Proposition 1. This concludes the proof. $\square$

With the help of Proposition 2, we can analyze the average normalized distortion $\mathbb{E}[\overline{D(\mathbf{X}, \mathbf{Y})}]$ by analyzing the individual expected distortions $\mathbb{E}[D_t(X_t, Y_t)]$ for any $t \in \mathbb{N}^+$ as the average normalized distortion $\mathbb{E}[\overline{D(\mathbf{X}, \mathbf{Y})}]$ is just the limit of the average of the expected distortions $\mathbb{E}[D_t(X_t, Y_t)]$. This will be useful in the proof of Theorem 1.

2) *A Probabilistic Tool:* A probabilistic tool that we will be using to prepare for the achievability proof is an extension of the weak law of large numbers (WLLN) given as follows.

Lemma 2: Let $\mathbf{X} = \{X_t\}_{t=1}^\infty$ be a sequence of real-valued i.n.i.d. RVs with a bounded variance sequence $\{\text{Var}[X_t]\}_{t=1}^\infty$. Let $S_N = X_1 + \dots + X_N$. Then, for any $\epsilon > 0$,

$$\lim_{N \rightarrow \infty} \Pr \left(\left| \frac{S_N}{N} - \frac{\sum_{t=1}^N \mathbb{E}[X_t]}{N} \right| < \epsilon \right) = 1. \quad (18)$$

Proof: For any $N \in \mathbb{N}^+$, let us see $\frac{S_N}{N}$ as an RV. Then, we have that

$$\begin{aligned}\text{Var} \left[\frac{S_N}{N} \right] &= \frac{\text{Var}[S_N]}{N^2} = \frac{\sum_{t=1}^N \text{Var}[X_t]}{N^2} \\ &\leq \frac{N \sup_t \{\text{Var}[X_t]\}}{N^2} \\ &= \frac{\sup_t \{\text{Var}[X_t]\}}{N} < \infty, \quad (19)\end{aligned}$$

which means that $\frac{S_N}{N}$ has finite variance $\text{Var}[\frac{S_N}{N}]$. When $\text{Var}[\frac{S_N}{N}] = 0$, $\forall 1 \leq t \leq N$,

$$\begin{aligned}\text{Var}[X_t] &= \mathbb{E}[(X_t - \mathbb{E}[X_t])^2] = 0 \\ \Rightarrow X_t &= \mathbb{E}[X_t] \text{ a.s. } (P_{X_t}).\end{aligned}$$

Consequently, for any $N \in \mathbb{N}^+$ such that $\text{Var}[\frac{S_N}{N}] = 0$,

$$\begin{aligned}\Pr \left(\frac{S_N}{N} = \frac{\sum_{t=1}^N \mathbb{E}[X_t]}{N} \right) \\ &= \Pr \left(\frac{\sum_{t=1}^N X_t}{N} = \frac{\sum_{t=1}^N \mathbb{E}[X_t]}{N} \right) = 1 \\ \Rightarrow \forall \epsilon > 0, \quad \Pr \left(\left| \frac{S_N}{N} - \frac{\sum_{t=1}^N \mathbb{E}[X_t]}{N} \right| < \epsilon \right) &= 1. \quad (20)\end{aligned}$$

By taking limits on both sides of (20), (18) is confirmed.

When $\text{Var}[\frac{S_N}{N}] > 0$, for any $\epsilon > 0$, let

$$k = \frac{\epsilon}{\sqrt{\text{Var}[\frac{S_N}{N}]}}.$$

Then, by Lemma 1, we write

$$\begin{aligned}\Pr \left(\left| \frac{S_N}{N} - \mathbb{E} \left[\frac{S_N}{N} \right] \right| \geq k \sqrt{\text{Var} \left[\frac{S_N}{N} \right]} \right) &\leq \frac{1}{k^2} \\ \Pr \left(\left| \frac{S_N}{N} - \frac{\sum_{t=1}^N \mathbb{E}[X_t]}{N} \right| \geq \epsilon \right) \\ &\leq \frac{\text{Var}[\frac{S_N}{N}]}{\epsilon^2} \leq \frac{\sup_t \{\text{Var}[X_t]\}}{N \epsilon^2} = O \left(\frac{1}{N} \right). \quad (21)\end{aligned}$$

Therefore,

$$\lim_{N \rightarrow \infty} \Pr \left(\left| \frac{S_N}{N} - \frac{\sum_{t=1}^N \mathbb{E}[X_t]}{N} \right| < \epsilon \right) = 1, \quad (22)$$

which concludes the proof. $\square$

This result is indeed known as it is a direct consequence of Chebyshev's inequality [43].

3) *The Information-Theoretic Tools:* Now, we are ready to develop the main tools that will be used in the proof.

Proposition 3 (i.n.i.d.-AEP): Let $\mathbf{X} = \{X_t\}_{t=1}^\infty$ be a sequence of i.n.i.d. RVs that take on values from finite alphabets $\{\mathcal{X}_t\}_{t=1}^\infty$ with a bounded cardinality sequence $\{|\mathcal{X}_t|\}_{t=1}^\infty$ and probability distributions $\{P_{X_t}\}_{t=1}^\infty$. Then, $\forall \epsilon > 0$,

$$\begin{aligned}\lim_{N \rightarrow \infty} \Pr \left(\left| -\frac{1}{N} \log_2 P_{X_1^N}(X_1, \dots, X_N) \right. \right. \\ \left. \left. - \frac{1}{N} H(X_1) \right| < \epsilon \right) = 1. \quad (23)\end{aligned}$$

Proof: For any $t \in \mathbb{N}^+$, we consider $P_{X_t} : \mathcal{X}_t \rightarrow [0, 1]$ as a function. Then, $\forall t \in \mathbb{N}^+$, $-\log_2 P_{X_t} : \mathcal{X}_t \rightarrow [0, \infty]$ is a function as well. Since functions of independent RVs are also independent RVs, the sequence of RVs $(-\log_2 P_{X_1}(X_1),$

$-\log_2 P_{X_2}(X_2), \dots$ is i.n.i.d. Since $\lim_{x \rightarrow 0} x \log_2 x = 0$, and

$$\begin{aligned} \lim_{x \rightarrow 0} x (\log_2 x)^2 &= \lim_{x \rightarrow 0} \frac{(\log_2 x)^2}{\frac{1}{x}} = \lim_{x \rightarrow 0} \frac{2 \log_2 x \cdot \frac{1}{x}}{-\frac{1}{x^2}} \\ &= \lim_{x \rightarrow 0} \frac{-2x \log_2 x}{\ln 2} = 0, \end{aligned} \quad (24)$$

we define the values of functions $g_1(x) = x \log_2 x$, $g_2(x) = x(\log_2 x)^2$ at point $x = 0$ to be 0, i.e., $g_1(0) = g_2(0) = 0$. Now, we analyze function g_2 . We write

$$\begin{aligned} g_2'(x) &= (\log_2 x)^2 + \frac{2 \log_2 x}{\ln 2} = 0 \\ \Rightarrow x &= 1 \quad \text{or} \quad x = \frac{1}{e^2}, \end{aligned}$$

which suggests that

$$\max_{x \in [0, 1]} g_2(x) = g_2\left(\frac{1}{e^2}\right) = \frac{4(\log_2 e)^2}{e^2}, \quad (25)$$

where we note that $g_2(1) = 0$ and $g_2(x)$ is increasing on $[0, \frac{1}{e^2}]$ and decreasing on $[\frac{1}{e^2}, 1]$.

As the cardinality sequence $\{|\mathcal{X}_t|\}_{t=1}^\infty$ is bounded, we can consider RV X with alphabet $\mathcal{X}$ such that $|\mathcal{X}| = \sup_t \{|\mathcal{X}_t|\}$. Then, the variance of RV $-\log_2 P_X(X)$ follows that

$$\begin{aligned} \text{Var}[-\log_2 P_X(X)] &= \mathbb{E} \left[(-\log_2 P_X(X))^2 \right] - (\mathbb{E}[-\log_2 P_X(X)])^2 \\ &\leq \mathbb{E} \left[(-\log_2 P_X(X))^2 \right] = \sum_{x \in \mathcal{X}} P_X(x) (\log_2 P_X(x))^2 \\ &\leq |\mathcal{X}| \frac{4(\log_2 e)^2}{e^2} < \infty, \end{aligned} \quad (26)$$

where the last inequality is obtained following (25). Consequently, following similar reasoning, for any $t \in \mathbb{N}^+$,

$$\text{Var}[-\log_2 P_{X_t}(X_t)] \leq |\mathcal{X}| \frac{4(\log_2 e)^2}{e^2}, \quad (27)$$

which means that the variance sequence $\{\text{Var}[-\log_2 P_{X_t}(X_t)]\}_{t=1}^\infty$ is bounded with an upper bound $|\mathcal{X}| \frac{4(\log_2 e)^2}{e^2}$.

For any $N \in \mathbb{N}^+$, by thinking of $-\frac{1}{N} \log_2 P_{X_1^N} : \mathcal{X}_1 \times \dots \times \mathcal{X}_N \rightarrow [0, \infty]$ as a function where $\forall x_1^N \in \mathcal{X}_1 \times \dots \times \mathcal{X}_N$,

$$-\frac{1}{N} \log_2 P_{X_1^N}(x_1^N) = -\frac{1}{N} \sum_{t=1}^N \log_2 P_{X_t}(x_t), \quad (28)$$

$-\frac{1}{N} \log_2 P_{X_1^N}(X_1, \dots, X_N)$ can be recognized as an RV $\frac{S_N}{N}$ where

$$S_N = (-\log_2 P_{X_1}(X_1)) + \dots + (-\log_2 P_{X_N}(X_N)). \quad (29)$$

Hence, by Lemma 2, (26), and (27), $\forall \epsilon > 0$,

$$\begin{aligned} \lim_{N \rightarrow \infty} \Pr \left(\left| -\frac{1}{N} \log_2 P_{X_1^N}(X_1, \dots, X_N) - \frac{1}{N} H(X_1^N) \right| < \epsilon \right) \\ &= \lim_{N \rightarrow \infty} \Pr \left(\left| -\frac{1}{N} \sum_{t=1}^N \log_2 P_{X_t}(X_t) - \frac{1}{N} \sum_{t=1}^N H(X_t) \right| < \epsilon \right) \\ &= 1, \end{aligned} \quad (30)$$

which completes the proof. $\square$

This result is generalized from the original Asymptotic Equipartition Property (AEP) [47] for i.i.d. random sequences. We call Proposition 3 *i.n.i.d.-AEP* because it considers a sequence of i.n.i.d. RVs. Following this result, we define the *i.n.i.d.-typical set* as follows.

Definition 2 (i.n.i.d.-typical set): Let $\mathbf{X} = \{X_t\}_{t=1}^\infty$ be a sequence of i.n.i.d. RVs that take on values from finite alphabets $\{\mathcal{X}_t\}_{t=1}^\infty$ with a bounded cardinality sequence $\{|\mathcal{X}_t|\}_{t=1}^\infty$ and probability distributions $\{P_{X_t}\}_{t=1}^\infty$. For any $\epsilon > 0$, the i.n.i.d.-typical set $A_\epsilon^{(N)}$ with respect to $P_{X_1^N}$ is the set of sequences $x_1^N = (x_1, \dots, x_N) \in \mathcal{X}_1 \times \dots \times \mathcal{X}_N$ with property

$$2^{-N(\frac{1}{N} H(X_1^N) + \epsilon)} \leq P_{X_1^N}(x_1^N) \leq 2^{-N(\frac{1}{N} H(X_1^N) - \epsilon)}. \quad (31)$$

Based on Proposition 3, we give some basic facts about the i.n.i.d.-typical set $A_\epsilon^{(N)}$.

Proposition 4: The i.n.i.d.-typical set $A_\epsilon^{(N)}$ has the following properties.

1) If $x_1^N \in A_\epsilon^{(N)}$, then

$$\begin{aligned} \frac{1}{N} H(X_1^N) - \epsilon &\leq -\frac{1}{N} \log_2 P_{X_1^N}(x_1, \dots, x_N) \\ &\leq \frac{1}{N} H(X_1^N) + \epsilon; \end{aligned}$$

2) $\lim_{N \rightarrow \infty} \Pr(A_\epsilon^{(N)}) = 1$;

3) $\overline{H(\mathbf{X})} - \epsilon \leq \lim_{N \rightarrow \infty} \frac{1}{N} \log_2(|A_\epsilon^{(N)}|) \leq \overline{H(\mathbf{X})} + \epsilon$.

Proof: The proof of 1) is straightforward by conducting operation $-\frac{1}{N} \log_2(\cdot)$ on all sides of the inequalities in (31). To prove 2), by Proposition 3, as $N \rightarrow \infty$,

$$\begin{aligned} -\frac{1}{N} \log_2 P_{X_1^N}(X_1, \dots, X_N) &\rightarrow \frac{1}{N} H(X_1^N) \quad \text{in probability} \\ \Rightarrow P_{X_1^N}(X_1, \dots, X_N) &\rightarrow 2^{-N(\frac{1}{N} H(X_1^N))} \quad \text{in probability.} \end{aligned} \quad (32)$$

As $2^{-N(\frac{1}{N} H(X_1^N) + \epsilon)} \leq 2^{-N(\frac{1}{N} H(X_1^N))} \leq 2^{-N(\frac{1}{N} H(X_1^N) - \epsilon)}$, by Definition 2, the probability of the event $\{X_1^N = (X_1, \dots, X_N) \in A_\epsilon^{(N)}\}$ tends to 1. Therefore, $\lim_{N \rightarrow \infty} \Pr(A_\epsilon^{(N)}) = 1$.

To prove 3), following Definition 2, we write

$$\begin{aligned} \sum_{x_1^N \in A_\epsilon^{(N)}} 2^{-N(\frac{1}{N} H(X_1^N) + \epsilon)} \\ &\leq \sum_{x_1^N \in A_\epsilon^{(N)}} P_{X_1^N}(x_1^N) \\ &\leq \sum_{x_1^N \in A_\epsilon^{(N)}} 2^{-N(\frac{1}{N} H(X_1^N) - \epsilon)} \end{aligned} \quad (33)$$

$$\begin{aligned} |A_\epsilon^{(N)}| 2^{-N(\frac{1}{N} H(X_1^N) + \epsilon)} \\ &\leq \sum_{x_1^N \in A_\epsilon^{(N)}} P_{X_1^N}(x_1^N) \\ &\leq |A_\epsilon^{(N)}| 2^{-N(\frac{1}{N} H(X_1^N) - \epsilon)}, \end{aligned} \quad (34)$$

where (33) holds because of (31). From 2), we have that

$$\lim_{N \rightarrow \infty} \sum_{x_1^N \in A_\epsilon^{(N)}} P_{X_1^N}(x_1^N) = 1.$$

Then,

$$\begin{aligned} \lim_{N \rightarrow \infty} |A_\epsilon^{(N)}| 2^{-N(\frac{1}{N} H(X_1^N) + \epsilon)} &\leq 1 \\ &\leq \lim_{N \rightarrow \infty} |A_\epsilon^{(N)}| 2^{-N(\frac{1}{N} H(X_1^N) - \epsilon)} \end{aligned} \quad (35)$$

$$\begin{aligned} \lim_{N \rightarrow \infty} \frac{1}{N} H(X_1^N) - \epsilon &\leq \lim_{N \rightarrow \infty} \frac{1}{N} \log_2 \left(|A_\epsilon^{(N)}| \right) \\ &\leq \lim_{N \rightarrow \infty} \frac{1}{N} H(X_1^N) + \epsilon \end{aligned} \quad (36)$$

$$\overline{H(\mathbf{X})} - \epsilon \leq \lim_{N \rightarrow \infty} \frac{1}{N} \log_2 \left(|A_\epsilon^{(N)}| \right) \leq \overline{H(\mathbf{X})} + \epsilon, \quad (37)$$

which concludes the proof. $\square$

We call the sequences in the i.n.i.d.-typical set *i.n.i.d.-typical sequences*. Proposition 4 shows that the probability that i.n.i.d.-typical sequences are sampled tends to 1 in N , the i.n.i.d.-typical sequences have similar probabilities, and the number of i.n.i.d.-typical sequences is about $2^{H(X_1^N)}$.

We then define the *i.n.i.d.-jointly typical set* as follows.

Definition 3 (i.n.i.d.-jointly typical set): Let $\mathbf{X} = \{X_t\}_{t=1}^\infty$, $\mathbf{U} = \{U_t\}_{t=1}^\infty$ be two sequences of i.n.i.d. RVs that take on values from finite alphabets $\{\mathcal{X}_t\}_{t=1}^\infty, \{\mathcal{U}_t\}_{t=1}^\infty$ with bounded cardinality sequences $\{|\mathcal{X}_t|\}_{t=1}^\infty, \{|\mathcal{U}_t|\}_{t=1}^\infty$ and probability distributions $\{P_{X_t}\}_{t=1}^\infty, \{P_{U_t}\}_{t=1}^\infty$, respectively. Let the sequence of RV pairs $(\mathbf{X}, \mathbf{U}) = \{(X_t, U_t)\}_{t=1}^\infty$ be i.n.i.d., i.e., $P_{\mathbf{X}, \mathbf{U}} = \otimes_t P_{X_t, U_t}$. Then, $\forall \epsilon > 0$, the i.n.i.d.-jointly typical set $J_\epsilon^{(N)}$ with respect to the probability distribution $P_{X_1^N, U_1^N}$ is the set of sequences $(x_1^N, u_1^N) \in \mathcal{X}_1^N \times \mathcal{U}_1^N$ such that

$$\left| -\frac{1}{N} \log_2 P_{X_1^N}(x_1, \dots, x_N) - \frac{1}{N} H(X_1^N) \right| < \epsilon, \quad (38)$$

$$\left| -\frac{1}{N} \log_2 P_{U_1^N}(u_1, \dots, u_N) - \frac{1}{N} H(U_1^N) \right| < \epsilon, \quad (39)$$

$$\left| -\frac{1}{N} \log_2 P_{X_1^N, U_1^N}(x_1^N, u_1^N) - \frac{1}{N} H(X_1^N, U_1^N) \right| < \epsilon. \quad (40)$$

The *i.n.i.d.-joint AEP* is given in the following proposition.

Proposition 5 (i.n.i.d.-joint AEP): The i.n.i.d.-jointly typical set $J_\epsilon^{(N)}$ has the following properties.

- 1) $\lim_{N \rightarrow \infty} \Pr \left((X_1^N, U_1^N) \in J_\epsilon^{(N)} \right) = 1.$
- 2) $\overline{H(\mathbf{X}, \mathbf{U})} - \epsilon \leq \lim_{N \rightarrow \infty} \frac{1}{N} \log_2(|J_\epsilon^{(N)}|) \leq \overline{H(\mathbf{X}, \mathbf{U})} + \epsilon.$
- 3) If $(\ddot{X}_1^N, \ddot{U}_1^N) \sim P_{X_1^N} P_{U_1^N}$, i.e., $\forall x_1^N \in \mathcal{X}_1^N, u_1^N \in \mathcal{U}_1^N, P_{\ddot{X}_1^N, \ddot{U}_1^N}(x_1^N, u_1^N) = P_{\ddot{X}_1^N}(x_1^N) P_{\ddot{U}_1^N}(u_1^N)$, then $\overline{I(\mathbf{X}; \mathbf{U})} - 3\epsilon \leq \lim_{N \rightarrow \infty} -\frac{1}{N} \log_2 \Pr((\ddot{X}_1^N, \ddot{U}_1^N) \in J_\epsilon^{(N)}) \leq \overline{I(\mathbf{X}; \mathbf{U})} + 3\epsilon.$

Proof: First, by Proposition 3, we have that

$$\lim_{N \rightarrow \infty} \Pr \left(\left| -\frac{1}{N} \log_2 P_{X_1^N}(X_1, \dots, X_N) \right| < \epsilon \right) = 1,$$

$$\left| -\frac{1}{N} H(X_1^N) \right| < \epsilon = 1, \quad (41)$$

$$\begin{aligned} \lim_{N \rightarrow \infty} \Pr \left(\left| -\frac{1}{N} \log_2 P_{U_1^N}(U_1, \dots, U_N) \right| < \epsilon \right) &= 1, \\ \left| -\frac{1}{N} H(U_1^N) \right| < \epsilon &= 1, \end{aligned} \quad (42)$$

$$\begin{aligned} \lim_{N \rightarrow \infty} \Pr \left(\left| -\frac{1}{N} \log_2 P_{X_1^N, U_1^N}(X_1^N, U_1^N) \right| < \epsilon \right) &= 1, \\ \left| -\frac{1}{N} H(X_1^N, U_1^N) \right| < \epsilon &= 1, \end{aligned} \quad (43)$$

where the third equality is obtained by combining $\mathbf{X}$ and $\mathbf{U}$ to be one sequence of i.n.i.d. RVs $(\mathbf{X}, \mathbf{U}) = \{(X_t, U_t)\}_t$. Then,

$$\begin{aligned} \lim_{N \rightarrow \infty} \Pr \left(\left| -\frac{1}{N} \log_2 P_{X_1^N}(X_1, \dots, X_N) \right| < \epsilon \right) &= 1, \\ \left| -\frac{1}{N} H(X_1^N) \right| < \epsilon &= 1, \end{aligned} \quad (44)$$

$$\begin{aligned} \lim_{N \rightarrow \infty} \Pr \left(\left| -\frac{1}{N} \log_2 P_{U_1^N}(U_1, \dots, U_N) \right| < \epsilon \right) &= 1, \\ \left| -\frac{1}{N} H(U_1^N) \right| < \epsilon &= 1, \end{aligned} \quad (45)$$

$$\begin{aligned} \lim_{N \rightarrow \infty} \Pr \left(\left| -\frac{1}{N} \log_2 P_{X_1^N, U_1^N}(X_1^N, U_1^N) \right| < \epsilon \right) &= 1, \\ \left| -\frac{1}{N} H(X_1^N, U_1^N) \right| < \epsilon &= 1. \end{aligned} \quad (46)$$

Then, the probability of the union of the events in (44), (45), and (46) also tends to 0. Therefore,

$$\lim_{N \rightarrow \infty} \Pr \left((X_1^N, U_1^N) \in J_\epsilon^{(N)} \right) = 1 - 0 = 1. \quad (47)$$

To prove 2), by (40), we have that for any $(x_1^N, u_1^N) \in J_\epsilon^{(N)}$, (48), shown at the bottom of the next page holds. By property 1), we have

$$\lim_{N \rightarrow \infty} \sum_{(x_1^N, u_1^N) \in J_\epsilon^{(N)}} P_{X_1^N, U_1^N}(x_1^N, u_1^N) = 1.$$

Consequently,

$$\begin{aligned} \lim_{N \rightarrow \infty} |J_\epsilon^{(N)}| 2^{-N(\frac{1}{N} H(X_1^N, U_1^N) + \epsilon)} &\leq 1 \\ &\leq \lim_{N \rightarrow \infty} |J_\epsilon^{(N)}| 2^{-N(\frac{1}{N} H(X_1^N, U_1^N) - \epsilon)} \\ \overline{H(\mathbf{X}, \mathbf{U})} - \epsilon &\leq \lim_{N \rightarrow \infty} \frac{1}{N} \log_2 \left(|J_\epsilon^{(N)}| \right) \leq \overline{H(\mathbf{X}, \mathbf{U})} + \epsilon. \end{aligned} \quad (49)$$

To prove 3), suppose that for any $N \in \mathbb{N}^+$, the random vectors $\ddot{X}_1^N$ and $\ddot{U}_1^N$ are independent of each other and have the same marginals as X_1^N, U_1^N , respectively. Then,

$$\begin{aligned} \Pr \left((\ddot{X}_1^N, \ddot{U}_1^N) \in J_\epsilon^{(N)} \right) &= \sum_{(x_1^N, u_1^N) \in J_\epsilon^{(N)}} P_{X_1^N}(x_1^N) P_{U_1^N}(u_1^N). \end{aligned} \quad (50)$$

By (38), (39) and property 2), we have

$$\begin{aligned}
& \lim_{N \rightarrow \infty} \sum_{(x_1^N, u_1^N) \in J_\epsilon^{(N)}} P_{X_1^N}(x_1^N) P_{U_1^N}(u_1^N) \\
& \leq \lim_{N \rightarrow \infty} |J_\epsilon^{(N)}| 2^{-N(\frac{1}{N}H(X_1^N) - \epsilon)} 2^{-N(\frac{1}{N}H(U_1^N) - \epsilon)} \\
& \leq \lim_{N \rightarrow \infty} 2^{N(\frac{1}{N}H(X_1^N, U_1^N) + \epsilon)} 2^{-N(\frac{1}{N}H(X_1^N) - \epsilon)} \\
& \quad 2^{-N(\frac{1}{N}H(U_1^N) - \epsilon)} \\
& = \lim_{N \rightarrow \infty} 2^{-N(\frac{1}{N}I(X_1^N; U_1^N) - 3\epsilon)} \\
& = 2^{-N(\overline{I(X; U)} - 3\epsilon)}. \tag{51}
\end{aligned}$$

Similarly, we also have

$$\begin{aligned}
& \lim_{N \rightarrow \infty} \sum_{(x_1^N, u_1^N) \in J_\epsilon^{(N)}} P_{X_1^N}(x_1^N) P_{U_1^N}(u_1^N) \\
& \geq \lim_{N \rightarrow \infty} |J_\epsilon^{(N)}| 2^{-N(\frac{1}{N}H(X_1^N) + \epsilon)} 2^{-N(\frac{1}{N}H(U_1^N) + \epsilon)} \\
& \geq \lim_{N \rightarrow \infty} 2^{N(\frac{1}{N}H(X_1^N, U_1^N) - \epsilon)} 2^{-N(\frac{1}{N}H(X_1^N) + \epsilon)} \\
& \quad 2^{-N(\frac{1}{N}H(U_1^N) + \epsilon)} \\
& = \lim_{N \rightarrow \infty} 2^{-N(\frac{1}{N}I(X_1^N; U_1^N) + 3\epsilon)} \\
& = 2^{-N(\overline{I(X; U)} + 3\epsilon)}. \tag{52}
\end{aligned}$$

Thus, by (51), (52) and (50),

$$\begin{aligned}
\overline{I(X; U)} - 3\epsilon & \leq \lim_{N \rightarrow \infty} -\frac{1}{N} \log_2 \Pr \left((\ddot{X}_1^N, \ddot{U}_1^N) \in J_\epsilon^{(N)} \right) \\
& \leq \overline{I(X; U)} + 3\epsilon,
\end{aligned}$$

which concludes the proof. $\square$

We call the sequences in set $J_\epsilon^{(N)}$ *i.n.i.d.-jointly typical sequences*. Proposition 5 shows that the i.n.i.d.-jointly typical set $J_\epsilon^{(N)}$ has about $2^{H(X_1^N)}$ and $2^{H(U_1^N)}$ i.n.i.d.-typical sequences for random vectors X_1^N and U_1^N , respectively. The number of i.n.i.d.-jointly typical sequences is about $2^{H(X_1^N, U_1^N)}$. The probability that a randomly independently sampled x_1^N, u_1^N subject to their marginal distributions $P_{X_1^N}, P_{U_1^N}$ is i.n.i.d.-jointly typical

is about $2^{-I(X_1^N, U_1^N)}$ and the probability that i.n.i.d.-jointly typical sequences are sampled subject to the joint distribution $P_{X_1^N, U_1^N}$ tends to $1/\ln N$.

By adding another condition to Definition 3 considering the normalized adaptive distortion function, we define the *i.n.i.d.-distortion typical set* as follows.

Definition 4 (i.n.i.d.-distortion typical set): Let $\mathbf{X} = \{X_t\}_{t=1}^\infty, \mathbf{U} = \{U_t\}_{t=1}^\infty$ be two sequences of i.n.i.d. RVs that take on values from finite alphabets $\{\mathcal{X}_t\}_{t=1}^\infty, \{\mathcal{U}_t\}_{t=1}^\infty$ with bounded cardinality sequences $\{|\mathcal{X}_t|\}_{t=1}^\infty, \{|\mathcal{U}_t|\}_{t=1}^\infty$ and probability distributions $\{P_{X_t}\}_{t=1}^\infty, \{P_{U_t}\}_{t=1}^\infty$, respectively. Let the sequence of RV pairs $(\mathbf{X}, \mathbf{U}) = \{(X_t, U_t)\}_{t=1}^\infty$ be i.n.i.d., i.e., $P_{\mathbf{X}, \mathbf{U}} = \otimes_t P_{X_t, U_t}$. Given a uniformly bounded normalized adaptive distortion function $\overline{D}(\cdot, \cdot) : \mathcal{X} \times \mathcal{U} \rightarrow [0, \infty), \forall \epsilon > 0$, the i.n.i.d.-distortion typical set $J_{D, \epsilon}^{(N)}$ with respect to the probability distribution $P_{X_1^N, U_1^N}$ is the set of sequences $(x_1^N, u_1^N) \in \mathcal{X}_1^N \times \mathcal{U}_1^N$ such that

$$\left| -\frac{1}{N} \log_2 P_{X_1^N}(x_1, \dots, x_N) - \frac{1}{N} H(X_1^N) \right| < \epsilon, \tag{53}$$

$$\left| -\frac{1}{N} \log_2 P_{U_1^N}(u_1, \dots, u_N) - \frac{1}{N} H(U_1^N) \right| < \epsilon, \tag{54}$$

$$\left| -\frac{1}{N} \log_2 P_{X_1^N, U_1^N}(x_1^N, u_1^N) - \frac{1}{N} H(X_1^N, U_1^N) \right| < \epsilon, \tag{55}$$

$$\left| \overline{D_1^N(x_1^N, u_1^N)} - \mathbb{E} \left[\overline{D_1^N(X_1^N, U_1^N)} \right] \right| < \epsilon. \tag{56}$$

Apparently, $J_{D, \epsilon}^{(N)} \subseteq J_\epsilon^{(N)}$. In fact, these two sets are almost the same as $N \rightarrow \infty$.

Proposition 6 (i.n.i.d.-distortion AEP): The i.n.i.d.-distortion typical set $J_{D, \epsilon}^{(N)}$ shares the same properties with the i.n.i.d.-jointly typical set $J_\epsilon^{(N)}$ as $N \rightarrow \infty$.

Proof: We only prove that

$$\lim_{N \rightarrow \infty} \Pr \left((X_1^N, U_1^N) \in J_{D, \epsilon}^{(N)} \right) = 1.$$

With this result proved, proofs of the other properties follow the same procedures as the proof of Proposition 5 for $J_\epsilon^{(N)}$.

For any $t \in \mathbb{N}^+$, $D_t(\cdot, \cdot) : \mathcal{X}_t \times \mathcal{U}_t \rightarrow [0, \infty)$ is a function of RVs X_t, U_t . Since functions of independent RVs are also independent RVs, The random sequence $\{D_t(X_t, U_t)\}_{t=1}^N$ is

$$\begin{aligned}
& \left| -\frac{1}{N} \log_2 P_{X_1^N, U_1^N}(x_1^N, u_1^N) - \frac{1}{N} H(X_1^N, U_1^N) \right| < \epsilon \\
& \Rightarrow \frac{1}{N} H(X_1^N, U_1^N) - \epsilon < -\frac{1}{N} \log_2 P_{X_1^N, U_1^N}(x_1^N, u_1^N) < \frac{1}{N} H(X_1^N, U_1^N) + \epsilon \\
& \Rightarrow 2^{-N(\frac{1}{N}H(X_1^N, U_1^N) + \epsilon)} < P_{X_1^N, U_1^N}(x_1^N, u_1^N) < 2^{-N(\frac{1}{N}H(X_1^N, U_1^N) - \epsilon)} \\
& \Rightarrow \sum_{(x_1^N, u_1^N) \in J_\epsilon^{(N)}} 2^{-N(\frac{1}{N}H(X_1^N, U_1^N) + \epsilon)} < \sum_{(x_1^N, u_1^N) \in J_\epsilon^{(N)}} P_{X_1^N, U_1^N}(x_1^N, u_1^N) < \sum_{(x_1^N, u_1^N) \in J_\epsilon^{(N)}} 2^{-N(\frac{1}{N}H(X_1^N, U_1^N) - \epsilon)} \\
& \Rightarrow |J_\epsilon^{(N)}| 2^{-N(\frac{1}{N}H(X_1^N, U_1^N) + \epsilon)} < \sum_{(x_1^N, u_1^N) \in J_\epsilon^{(N)}} P_{X_1^N, U_1^N}(x_1^N, u_1^N) < |J_\epsilon^{(N)}| 2^{-N(\frac{1}{N}H(X_1^N, U_1^N) - \epsilon)}. \tag{48}
\end{aligned}$$

i.n.i.d. Then, we write $\forall t \in \mathbb{N}^+$,

$$\begin{aligned} & \mathbb{E}[D_t(X_t, U_t)] \\ &= \sum_{x_t \in \mathcal{X}_t, u_t \in \mathcal{U}_t} P_{X_t, U_t}(x_t, u_t) D_t(x_t, u_t) \leq \gamma_{\max}, \\ & \mathbb{E}[(D_t(X_t, U_t))^2] \\ &= \sum_{x_t, u_t} P_{X_t, U_t}(x_t, u_t) (D_t(X_t, U_t))^2 \leq \gamma_{\max}^2, \end{aligned} \quad (57)$$

where $\gamma_{\max}$ is the uniform bound of $\overline{D(\cdot, \cdot)}$. Therefore,

$$\begin{aligned} \text{Var}[D_t(X_t, U_t)] &= \mathbb{E}[(D_t(X_t, U_t))^2] - (\mathbb{E}[D_t(X_t, U_t)])^2 \\ &\leq \gamma_{\max}^2 < \infty. \end{aligned} \quad (58)$$

Let $S_N = D_1^N(X_1^N, U_1^N) = \sum_{t=1}^N D_t(X_t, U_t)$. By Proposition 3 and Proposition 1, we have that $\forall \epsilon > 0$,

$$\begin{aligned} & \lim_{N \rightarrow \infty} \Pr \left(\left| \frac{S_N}{N} - \frac{1}{N} \sum_{t=1}^N \mathbb{E}[D_t(X_t, U_t)] \right| < \epsilon \right) \\ &= \lim_{N \rightarrow \infty} \Pr \left(\left| \overline{D_1^N(X_1^N, U_1^N)} - \mathbb{E}[\overline{D_1^N(X_1^N, U_1^N)}] \right| < \epsilon \right) \\ &= 1. \end{aligned} \quad (59)$$

Because the same also happens for (53), (54), and (55), following the steps of the proof of Proposition 5, we have $\lim_{N \rightarrow \infty} \Pr((X_1^N, U_1^N) \in J_{D, \epsilon}^{(N)}) = 1$. $\square$

The sequences in set $J_{D, \epsilon}^{(N)}$ are called *i.n.i.d.-distortion typical sequences*.

B. Proof of Achievability

To prove the achievability of the bound, we construct a special random binning as follows. First, fix conditional probability distributions $\{P_{U_t|X_t}\}_t$ and functions $\{f_t\}_t$ such that $(\{P_{U_t|X_t}\}_{t=1}^\infty, \{f_t\}_{t=1}^\infty) \in \mathcal{P}_d$. Then, for any small number $\epsilon > 0$, $\forall N \in \mathbb{N}^+$, randomly generate a codebook $\mathcal{C}^{(N)}$ with $2^{N(\frac{1}{N}I(\hat{Y}_1^N; U_1^N) - 4\epsilon)}$ i.n.i.d. codewords $U_1^N \sim \otimes_{t=1}^N P_{U_t}$, where $\forall t \in \mathbb{N}^+, u_t \in \mathcal{U}_t, P_{U_t}(u_t) = \sum_{x_t \in \mathcal{X}_t} P_{X_t}(x_t) P_{U_t|X_t}(u_t|x_t)$. The codewords in the codebook $\mathcal{C}^{(N)}$ are indexed by $\{1, \dots, 2^{N(\frac{1}{N}I(\hat{Y}_1^N; U_1^N) - 4\epsilon)}\}$ and are then randomly distributed into $2^{N(\frac{I(\hat{Y}_1^N; U_1^N) - I(X_1^N; U_1^N)}{N} - 8\epsilon)}$ bins $\{\mathcal{C}_j^{(N)}\}$ which are indexed by $\mathcal{M} = \{1, \dots, 2^{N(\frac{I(\hat{Y}_1^N; U_1^N) - I(X_1^N; U_1^N)}{N} - 8\epsilon)}\}$, where $\mathcal{M}$ is the alphabet of the secret message M and $j \in \mathcal{M}$. So, there are approximately $2^{N(\frac{1}{N}I(X_1^N; U_1^N) + 4\epsilon)}$ codewords in each bin $\mathcal{C}_j^{(N)}$.

Given a cover vector x_1^N generated with respect to the distribution $P_{X_1^N}$, an i.n.i.d.-distortion typical encoder is applied which finds the codeword u_1^N whose corresponding stego vector $y_1^N = (f_1(u_1, x_1), f_2(u_2, x_2), \dots, f_N(u_N, x_N))$ is i.n.i.d.-distortion typical with the cover vector x_1^N in the bin specified by the secret message $m \in \mathcal{M}$, i.e., it finds $u_1^N \in \mathcal{C}_m^{(N)}$, such

that $(x_1^N, y_1^N) \in J_{D, \epsilon}^{(N)}$. If there is no codeword u_1^N such that the corresponding y_1^N is i.n.i.d.-distortion typical with x_1^N , the encoder randomly selects a codeword $u_1^N \in \mathcal{C}_m^{(N)}$. By Definition 4, if x_1^N and y_1^N are i.n.i.d.-distortion typical, we have that

$$\begin{aligned} & \left| -\frac{1}{N} \log_2 P_{X_1^N, Y_1^N}(x_1^N, y_1^N) - \frac{1}{N} H(X_1^N, Y_1^N) \right| < \epsilon \\ \Rightarrow & \left| -\frac{1}{N} \log_2 P_{X_1^N, U_1^N}(x_1^N, u_1^N) - \frac{1}{N} H(X_1^N, U_1^N) \right| < \epsilon, \end{aligned}$$

as functions $\{f_t\}_t$ are fixed. Since the same also happens for the individual vectors x_1^N, u_1^N , the cover vector x_1^N and the codeword u_1^N are i.n.i.d.-jointly typical.

The stego $y_1^N = (f_1(u_1, x_1), f_2(u_2, x_2), \dots, f_N(u_N, x_N))$ is transmitted through the noisy non-stationary sequence of discrete memoryless channels and becomes $\hat{y}_1^N$. At the receiver side, an i.n.i.d.-jointly typical decoder is applied which first finds the codeword $\hat{u}_1^N$ that is i.n.i.d.-jointly typical with the channel output $\hat{y}_1^N$ in the whole codebook $\mathcal{C}^{(N)}$ and then obtain the corresponding bin index $\hat{m}$. If there is no codeword $\hat{u}_1^N \in \mathcal{C}^{(N)}$ that is i.n.i.d.-jointly typical with $\hat{y}_1^N$, the decoder randomly selects $\hat{u}_1^N$ and obtains its bin index.

Now, we analyze the performance of the above scheme. As the alphabet $\mathcal{M}$ is given, the maximum information that can be embedded is $H(M) = N(\frac{I(\hat{Y}_1^N; U_1^N) - I(X_1^N; U_1^N)}{N} - 8\epsilon)$ where M is uniformly distributed over $\mathcal{M}$. Then, we have that

$$\begin{aligned} \frac{H(M)}{N} &= \frac{I(\hat{Y}_1^N; U_1^N) - I(X_1^N; U_1^N)}{N} - 8\epsilon \\ \Rightarrow \lim_{N \rightarrow \infty} \frac{H(M)}{N} &= \lim_{N \rightarrow \infty} \frac{I(\hat{Y}_1^N; U_1^N) - I(X_1^N; U_1^N)}{N} - 8\epsilon \\ &= \overline{I(\hat{Y}; U)} - \overline{I(X; U)} - 8\epsilon. \end{aligned} \quad (60)$$

Therefore, as $\epsilon \rightarrow 0$, following the choices of conditional probability distributions $\{P_{U_t|X_t}\}_t$ and functions $\{f_t\}_t$, $\overline{I(\hat{Y}; U)} - \overline{I(X; U)}$ is achieved as N tends to infinity. It remains to show that the extraction error tends to 0 in N and the expected average normalized distortion of the scheme tends to be less than or equal to the design distortion d as N tends to infinity. Without loss of generality, we can assume that the message $m = 1$ is going to be communicated. There are four possible error events:

There is no codeword $u_1^N \in \mathcal{C}_m^{(N)}$ such that the corresponding y_1^N is i.n.i.d.-distortion typical with the cover vector x_1^N . We denote this event by E_1 . First, we note that $\Pr(X_1^N \in A_\epsilon^{(N)})$ tends to 1 in N by Proposition 4. Let $x_1^N \in A_\epsilon^{(N)}$ be a fixed cover vector. Then, the probability that a randomly chosen sample vector u_1^N whose corresponding y_1^N is not i.n.i.d.-distortion typical with x_1^N is given by

$$\begin{aligned} & \Pr((x_1^N, Y_1^N) \notin J_{D, \epsilon}^{(N)}) \\ &= \Pr((x_1^N, f_1^N(U_1^N, x_1^N)) \notin J_{D, \epsilon}^{(N)}) \\ &= 1 - \sum_{\hat{u}_1^N \in \mathcal{U}_1^{(N)}: (x_1^N, f_1^N(\hat{u}_1^N, x_1^N)) \in J_{D, \epsilon}^{(N)}} P_{U_1^N}(\hat{u}_1^N), \end{aligned} \quad (61)$$

where $f_1^N(\dot{u}_1^N, x_1^N) = (f_1(\dot{u}_1, x_1), f_2(\dot{u}_2, x_2), \dots, f_N(\dot{u}_N, x_N))$.

The bin $\mathcal{C}_1^{(N)}$ can be seen as a set containing around $2^{N(\frac{1}{N}I(X_1^N; U_1^N) + 4\epsilon)}$ randomly generated codewords subject to the probability distribution $P_{U_1^N}$. By the WLLN, as $N \rightarrow \infty$, the probability $\Pr(|\mathcal{C}_1^{(N)}| = 2^{N(\frac{1}{N}I(X_1^N; U_1^N) + 4\epsilon)})$ tends to 1. Therefore, (62), shown at the bottom of this page holds.

Since for any $(x_1^N, f_1^N(\dot{u}_1^N, x_1^N)) \in J_{D,\epsilon}^{(N)}$, the vectors x_1^N and $\dot{u}_1^N$ are i.n.i.d.-jointly typical, by Definition 3,

$$\begin{aligned} P_{U_1^N|X_1^N}(\dot{u}_1^N|x_1^N) &= \frac{P_{U_1^N, X_1^N}(\dot{u}_1^N, x_1^N)}{P_{X_1^N}(x_1^N)} \\ &= P_{U_1^N}(\dot{u}_1^N) \frac{P_{U_1^N, X_1^N}(\dot{u}_1^N, x_1^N)}{P_{U_1^N}(\dot{u}_1^N)P_{X_1^N}(x_1^N)} \\ &\leq P_{U_1^N}(\dot{u}_1^N) \frac{2^{-N(\frac{1}{N}H(U_1^N, X_1^N) - \epsilon)}}{2^{-N(\frac{1}{N}H(U_1^N) + \epsilon)} 2^{-N(\frac{1}{N}H(X_1^N) + \epsilon)}} \\ &= P_{U_1^N}(\dot{u}_1^N) 2^{N(\frac{1}{N}I(X_1^N; U_1^N) + 3\epsilon)} \\ &\Rightarrow P_{U_1^N}(\dot{u}_1^N) \geq P_{U_1^N|X_1^N}(\dot{u}_1^N|x_1^N) 2^{-N(\frac{1}{N}I(X_1^N; U_1^N) + 3\epsilon)}. \end{aligned} \quad (63)$$

Then, by substituting (63) into (62), we have (64) shown at the bottom of this page, where the second inequality is obtained by [47][Lemma 10.5.3] and the last line is by Proposition 6. So the probability of this error event tends to 0 in N .

The sequences y_1^N and $\hat{y}_1^N$ are not i.n.i.d.-jointly typical. We denote this error event by E_2 . The probability of this error event

tends to 0 in N by Proposition 5 as they are the input and output of a non-stationary sequence of discrete memoryless channels, respectively.

The sequences $\hat{y}_1^N$ and u_1^N are not i.n.i.d.-jointly typical. We denote this error event by E_3 . Because $\lim_{N \rightarrow \infty} \Pr(E_1) = 0$, u_1^N and x_1^N are i.n.i.d.-jointly typical with probability 1 as $N \rightarrow \infty$. Also, as $\lim_{N \rightarrow \infty} \Pr(E_2) = 0$, $y_1^N = (f_1(u_1, x_1), f_2(u_2, x_2), \dots, f_N(u_N, x_N))$ and $\hat{y}_1^N$ are i.n.i.d.-jointly typical with probability 1 as $N \rightarrow \infty$. Since $U_1^N \rightarrow (X_1^N, Y_1^N) \rightarrow \hat{Y}_1^N$ form a Markov chain, the probability that u_1^N and $\hat{y}_1^N$ are i.n.i.d.-jointly typical tends to 1 in N by the Markov Lemma [47], [48]. Therefore, the probability of the error event is negligible, i.e., $\lim_{N \rightarrow \infty} \Pr(E_3) = 0$.

There exists another $\hat{u}_1^N \neq u_1^N$ that is i.n.i.d.-jointly typical with $\hat{y}_1^N$, and the decoder output is $\hat{u}_1^N$, and its bin index $\hat{m}$. We denote this error event by E_4 . Since the codebook $\mathcal{C}^{(N)}$ is randomly generated and is independent of $\hat{y}_1^N$, by Proposition 5, as N tends to infinity, the probability that a random codeword in $\mathcal{C}^{(N)}$ that is i.n.i.d.-jointly typical with $\hat{y}_1^N$ is upper bounded by $2^{-N(I(\hat{Y}; U) - 3\epsilon)}$. Then,

$$\begin{aligned} \lim_{N \rightarrow \infty} \Pr(E_4) &\leq \lim_{N \rightarrow \infty} 2^{N(\frac{1}{N}I(\hat{Y}_1^N; U_1^N) - 4\epsilon)} 2^{-N(I(\hat{Y}; U) - 3\epsilon)} \\ &= \lim_{N \rightarrow \infty} 2^{-N\epsilon} = 0. \end{aligned} \quad (65)$$

Now, we have proved that $\forall k \in \{1, 2, 3, 4\}$, $\lim_{N \rightarrow \infty} \Pr(E_k) = 0$, which confirms that the extraction error indeed

$$\begin{aligned} \lim_{N \rightarrow \infty} \Pr(E_1) &= \lim_{N \rightarrow \infty} \sum_{x_1^N \in \mathcal{X}_1^N} P_{X_1^N}(x_1^N) \times \left(1 - \sum_{\dot{u}_1^N \in \mathcal{U}_1^N: (x_1^N, f_1^N(\dot{u}_1^N, x_1^N)) \in J_{D,\epsilon}^{(N)}} P_{U_1^N}(\dot{u}_1^N) \right) 2^{N(\frac{1}{N}I(X_1^N; U_1^N) + 4\epsilon)} \\ &= \lim_{N \rightarrow \infty} \sum_{x_1^N \in \mathcal{A}_\epsilon^{(N)}} P_{X_1^N}(x_1^N) \times \left(1 - \sum_{\dot{u}_1^N \in \mathcal{U}_1^N: (x_1^N, f_1^N(\dot{u}_1^N, x_1^N)) \in J_{D,\epsilon}^{(N)}} P_{U_1^N}(\dot{u}_1^N) \right) 2^{N(\frac{1}{N}I(X_1^N; U_1^N) + 4\epsilon)}. \end{aligned} \quad (62)$$

$$\begin{aligned} &\lim_{N \rightarrow \infty} \Pr(E_1) \\ &\leq \lim_{N \rightarrow \infty} \sum_{x_1^N \in \mathcal{A}_\epsilon^{(N)}} P_{X_1^N}(x_1^N) \left(1 - \sum_{\dot{u}_1^N \in \mathcal{U}_1^N: (x_1^N, f_1^N(\dot{u}_1^N, x_1^N)) \in J_{D,\epsilon}^{(N)}} P_{U_1^N|X_1^N}(\dot{u}_1^N|x_1^N) 2^{-N(\frac{1}{N}I(X_1^N; U_1^N) + 3\epsilon)} \right) 2^{N(\frac{1}{N}I(X_1^N; U_1^N) + 4\epsilon)} \\ &\leq \lim_{N \rightarrow \infty} \sum_{x_1^N \in \mathcal{A}_\epsilon^{(N)}} P_{X_1^N}(x_1^N) \left(1 - \sum_{\dot{u}_1^N \in \mathcal{U}_1^N: (x_1^N, f_1^N(\dot{u}_1^N, x_1^N)) \in J_{D,\epsilon}^{(N)}} P_{U_1^N|X_1^N}(\dot{u}_1^N|x_1^N) + e^{-2^{-N(\frac{1}{N}I(X_1^N; U_1^N) + 3\epsilon)}} 2^{N(\frac{1}{N}I(X_1^N; U_1^N) + 4\epsilon)} \right) \\ &= 1 + \lim_{N \rightarrow \infty} e^{-2^{-N\epsilon}} - \lim_{N \rightarrow \infty} \left(\sum_{x_1^N \in \mathcal{A}_\epsilon^{(N)}, \dot{u}_1^N \in \mathcal{U}_1^N: (x_1^N, f_1^N(\dot{u}_1^N, x_1^N)) \in J_{D,\epsilon}^{(N)}} P_{U_1^N|X_1^N}(\dot{u}_1^N|x_1^N) P_{X_1^N}(x_1^N) \right) = 0. \end{aligned} \quad (64)$$

tends to 0 in N . Moreover, since $\lim_{N \rightarrow \infty} \Pr(E_1) = 0$, the random sampled cover vector x_1^N and the stego vector y_1^N are i.n.i.d.-distortion typical with probability 1 as N tends to infinity. So, by Definition 4 and Proposition 6,

$$\begin{aligned} 1 &= \lim_{N \rightarrow \infty} \Pr \left(\overline{D_1^N(X_1^N, Y_1^N)} < \mathbb{E} \left[\overline{D_1^N(X_1^N, Y_1^N)} \right] + \epsilon \right) \\ &= \Pr \left(\overline{D(\mathbf{X}, \mathbf{Y})} < \mathbb{E} \left[\overline{D(\mathbf{X}, \mathbf{Y})} \right] + \epsilon \right). \end{aligned} \quad (66)$$

Following (5), $\mathbb{E}[\overline{D(\mathbf{X}, \mathbf{Y})}] \leq d$ is a constant. Consequently,

$$\Pr \left(\overline{D(\mathbf{X}, \mathbf{Y})} < d + \epsilon \right) = 1. \quad (67)$$

Then, as $\epsilon \rightarrow 0$, the expected average normalized distortion of the random binning scheme would be less than or equal to the design distortion d . Finally, by the arbitrariness of the choices of $\{P_{U_t|X_t}\}_t$ and $\{f_t\}_t$ such that $(\{P_{U_t|X_t}\}_{t=1}^\infty, \{f_t\}_{t=1}^\infty) \in \mathcal{P}_d$, we have that any rate $R < R_D(d)$ is achievable. This concludes the proof of achievability.

C. Proof of the Converse

We confine the problem to a smaller scenario. Suppose that the design distortion d is very large, e.g., $d \geq \gamma_{\max}$. Then, as d increases, the rate-distortion bound $R_D(d)$ will not change and will become flat because $\mathbb{E}[\overline{D(\mathbf{X}, \mathbf{Y})}] \leq d$ will always be true no matter how the probability distributions $\{P_{U_t|X_t}\}_{t=1}^\infty$ and functions $\{f_t\}_{t=1}^\infty$ vary. As the set $\mathcal{P}_d$ will not get smaller with d increasing, following (7), $R_D(d)$ is non-decreasing. Thus, we can define $d_{\max} \geq 0$ to be

$$d_{\max} \triangleq \inf_{R_D(d) \text{ is flat}} d. \quad (68)$$

We only consider the situation when $d \leq d_{\max}$ in this proof where we can hope that the average normalized distortion tends to d as $N \rightarrow \infty$ for optimal (R, d) pairs. When $d > d_{\max}$, we have that $R_D(d) = R_D(d_{\max})$. So, the pair $(R_D(d_{\max}), d_{\max})$ applies to this situation. To show the optimality of the bound (7), we need the following lemma.

Lemma 3: Let design distortion $d \leq d_{\max}$. There exists a sequence of non-negative real numbers $\{d_t\}_t$ such that $\overline{\sum_t d_t} = d$ and the rate-distortion bound (7) satisfies

$$R_D(d) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N R_{D_t}(d_t), \quad (69)$$

where for any $t \in \mathbb{N}^+$, $R_{D_t}(d_t)$ is defined as

$$R_{D_t}(d_t) \triangleq \sup_{(P_{U_t|X_t}, f_t) \in \mathcal{P}_{d_t}^{(1)}} \left(I(\hat{Y}_t; U_t) - I(X_t; U_t) \right), \quad (70)$$

with set $\mathcal{P}_{d_t}^{(1)}$ being defined as

$$\begin{aligned} \mathcal{P}_{d_t}^{(1)} \triangleq \left\{ P_{U_t|X_t}, f_t \mid f_t : \mathcal{U}_t \times \mathcal{X}_t \rightarrow \mathcal{Y}_t, Y_t = f_t(U_t, X_t); \right. \\ \left. \mathbb{E}[D_t(X_t, Y_t)] \leq d_t \right\}. \end{aligned} \quad (71)$$

Proof: For any sequence of non-negative real numbers $\{d_t\}_t$ such that $\overline{\sum_t d_t} = d$, if $\forall t \in \mathbb{N}^+$, $\mathbb{E}[D_t(X_t, Y_t)] \leq d_t$,

$$\begin{aligned} \mathbb{E}[\overline{D(\mathbf{X}, \mathbf{Y})}] &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \mathbb{E}[D_t(X_t, Y_t)] \\ &\leq \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N d_t = d. \end{aligned}$$

Consequently, for conditional distributions $\{P_{U_t|X_t}\}_{t=1}^\infty$ and functions $\{f_t\}_{t=1}^\infty$ such that $\forall t \in \mathbb{N}^+$, $(P_{U_t|X_t}, f_t) \in \mathcal{P}_{d_t}^{(1)}$, we have that $(\{P_{U_t|X_t}\}_{t=1}^\infty, \{f_t\}_{t=1}^\infty) \in \mathcal{P}_d$. As a result,

$$R_D(d) = \sup_{(\{P_{U_t|X_t}\}_{t=1}^\infty, \{f_t\}_{t=1}^\infty) \in \mathcal{P}_d} \overline{I(\hat{\mathbf{Y}}; \mathbf{U})} - \overline{I(\mathbf{X}; \mathbf{U})} \quad (72)$$

$$\geq \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \sup_{(P_{U_t|X_t}, f_t) \in \mathcal{P}_{d_t}^{(1)}} \left(I(\hat{Y}_t; U_t) - I(X_t; U_t) \right) \quad (73)$$

$$= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N R_{D_t}(d_t). \quad (74)$$

Indeed, the converse of (73) also holds. To show this, following the traditional information embedding theorem [38], $R_{D_t}(d_t)$ is the rate-distortion bound in the single-letter scenario where $\mathbb{E}[D_t(X_t, Y_t)] \leq d_t$ with respect to the optimal $(P_{U_t|X_t}, f_t) \in \mathcal{P}_{d_t}^{(1)}$, the cover probability distribution P_{X_t} , and the DMC W_t with transition probabilities $\{P_{\hat{Y}_t|Y_t}(\hat{y}_t|y_t)\}$. Then, for any $t \in \mathbb{N}^+$, when $\mathbb{E}[D_t(X_t, Y_t)] \leq d_t$,

$$\begin{aligned} I(\hat{Y}_t; U_t) - I(X_t; U_t) &\leq \sup_{(P_{U_t|X_t}, f_t) \in \mathcal{P}_{d_t}^{(1)}} \left(I(\hat{Y}_t; U_t) - I(X_t; U_t) \right) \\ &= R_{D_t}(d_t). \end{aligned} \quad (75)$$

If $\forall t, \mathbb{E}[D_t(X_t, Y_t)] \leq d_t$, by substituting (75) into (72),

$$\begin{aligned} \overline{I(\hat{\mathbf{Y}}; \mathbf{U})} - \overline{I(\mathbf{X}; \mathbf{U})} &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \left(I(\hat{Y}_t; U_t) - I(X_t; U_t) \right) \\ &\leq \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \sup_{(P_{U_t|X_t}, f_t) \in \mathcal{P}_{d_t}^{(1)}} \left(I(\hat{Y}_t; U_t) - I(X_t; U_t) \right) \\ &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N R_{D_t}(d_t). \end{aligned} \quad (76)$$

Let $(\{P_{U_t|X_t}^i\}_{t=1}^\infty, \{f_t^i\}_{t=1}^\infty)_{i=1}^\infty$ be a bound-achieving sequence of elements in $\mathcal{P}_d$. Take

$$d_t = \lim_{i \rightarrow \infty} \mathbb{E} [D_t(X_t, f_t^i(U_t^i, X_t))] \quad \forall t \in \mathbb{N}^+, \quad (77)$$

where for any $u_t \in \mathcal{U}_t$ and any $i \in \mathbb{N}^+$, $P_{U_t^i}(u_t) = \sum_{x_t \in \mathcal{X}_t} P_{X_t}(x_t) P_{U_t|X_t}^i(u_t|x_t)$. Then, $\overline{\sum_t d_t} = d$ by the definition of set $\mathcal{P}_d$ and the fact that $d \leq d_{\max} \leq \gamma_{\max}$. As i tends to infinity,

we have that $\lim_{i \rightarrow \infty} (P_{U_t|X_t}^i, f_t^i) \in \mathcal{P}_{d_t}^{(1)}$ for any $t \in \mathbb{N}^+$. Therefore, by substituting (77) into (76) and taking the supremum on all sides of (76),

$$\begin{aligned} R_D(d) &= \sup_{(\{P_{U_t|X_t}\}_{t=1}^\infty, \{f_t\}_{t=1}^\infty) \in \mathcal{P}_d} \overline{I(\hat{Y}; U)} - \overline{I(X; U)} \\ &\leq \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N R_{D_t}(d_t). \end{aligned} \quad (78)$$

This concludes the proof. $\square$

Lemma 3 indicates that the rate-distortion bound (7) can be decomposed into the limit of the normalized sum of a sequence of single-letter rate-distortion functions $\{R_{D_t}(d_t)\}_t$ with corresponding design distortions $\{d_t\}_t$ that satisfy $\sum_t d_t = d$.

Similar to Lemma 3, we have the following corollary.

Corollary 1: Let $d \leq d_{\max}$. There exists a sequence of non-negative real numbers $\{d_t\}_t$ such that $\sum_t d_t = d$ and

$$\begin{aligned} &\sup_{\{P_{U_t, X_t, Y_t, \hat{Y}_t}\}_{t=1}^\infty, \mathbb{E}[\overline{D(X, Y)}] \leq d} \overline{I(\hat{Y}; U)} - \overline{I(X; U)} \\ &= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \\ &\quad \times \sup_{P_{U_t, X_t, Y_t, \hat{Y}_t}, \mathbb{E}[D_t(X_t, Y_t)] \leq d_t} \left(I(\hat{Y}_t; U_t) - I(X_t; U_t) \right). \end{aligned} \quad (79)$$

We omit the proof of this result as it follows almost the same procedures of the proof of Lemma 3. We only make a note here that there are no functions $\{f_t\}_t$ in the setting of the corollary and the expectations $\mathbb{E}[\overline{D(X, Y)}]$ and $\{\mathbb{E}[D_t(X_t, Y_t)]\}_t$ only depend on the corresponding probability distributions. Corollary 1 suggests that the decomposition also holds for the supremum taken in the whole domain of the probability distributions with the absence of functions $\{f_t\}_t$.

There is another lemma that we need for the proof.

Lemma 4: For fixed cover distribution $P_X = \otimes_t P_{X_t}$, uniformly bounded normalized adaptive distortion function $\overline{D(\cdot, \cdot)} : \mathcal{X} \times \mathcal{Y} \rightarrow [0, \infty)$, and non-stationary sequence of discrete memoryless channels $\{W_t\}_t$ with transition probabilities $\{\{P_{\hat{Y}_t|Y_t}(\hat{y}_t|y_t)\}\}_t$, if $d \leq d_{\max}$,

$$\begin{aligned} &\sup_{(\{P_{U_t|X_t}\}_{t=1}^\infty, \{f_t\}_{t=1}^\infty) \in \mathcal{P}_d} \overline{I(\hat{Y}; U)} - \overline{I(X; U)} \\ &= \sup_{\{P_{U_t, X_t, Y_t, \hat{Y}_t}\}_{t=1}^\infty, \mathbb{E}[\overline{D(X, Y)}] \leq d} \overline{I(\hat{Y}; U)} - \overline{I(X; U)}. \end{aligned} \quad (80)$$

Proof: It is obvious that

$$R_D(d) \leq \sup_{\{P_{U_t, X_t, Y_t, \hat{Y}_t}\}_{t=1}^\infty, \mathbb{E}[\overline{D(X, Y)}] \leq d} \overline{I(\hat{Y}; U)} - \overline{I(X; U)}$$

as the functions $\{f_t\}_t$ can be seen as special conditional distributions. We now show the converse of the inequality. $\forall t \in \mathbb{N}^+$

and $\forall u_t \in \mathcal{U}_t, x_t \in \mathcal{X}_t, y_t \in \mathcal{Y}_t, \hat{y}_t \in \hat{\mathcal{Y}}_t$,

$$\begin{aligned} &P_{U_t, X_t, Y_t, \hat{Y}_t}(u_t, x_t, y_t, \hat{y}_t) \\ &= P_{X_t}(x_t) P_{U_t|X_t}(u_t|x_t) P_{Y_t|X_t, U_t}(y_t|x_t, u_t) P_{\hat{Y}_t|Y_t}(\hat{y}_t|y_t). \end{aligned} \quad (81)$$

Then, since distributions $\{P_{X_t}\}_{t=1}^\infty$ and $\{P_{\hat{Y}_t|Y_t}\}_{t=1}^\infty$ are fixed, $\exists \{d_t\}_t$ such that (82), (83), and (84), shown at the bottom of the next page hold, where (82) follows from Corollary 1, (83) follows from (81), and (84) is obtained by [38, Lemma 2], (73), and Lemma 3. $\square$

Lemma 4 shows that the supremum in (7) is indeed equal to the supremum taken in the whole domain of the probability distributions. So, if the optimality of the bound in the latter situation is confirmed, the bound in (7) is optimal.

With all preparations ready, we now prove the converse which generalizes the Gel'fand-Pinsker Theorem in the absence of a distortion constraint under the single-letter setting [46] to the case with an adaptive distortion function under the non-stationary memoryless setting.

Proof of the Converse Theorem Let a code with parameters $(N, H(M), \Delta)$ be defined by two mappings: the encoding mapping $F_E : \mathcal{X}_1^N \times \mathcal{M} \rightarrow \mathcal{Y}_1^N$ and the decoding mapping $F_D : \mathcal{Y}_1^N \rightarrow \mathcal{M}$ with expected average normalized distortion

$$\begin{aligned} \mathbb{E}_F \left[\overline{D_1^N(X_1^N, Y_1^N)} \right] &= \frac{1}{N} \mathbb{E} \left[\sum_{t=1}^N D_t(X_t, F_{E,t}(X_1^N, M)) \right] \\ &= \frac{1}{N} \sum_{m \in \mathcal{M}} \sum_{x_1^N \in \mathcal{X}_1^N} \sum_{t=1}^N P_M(m) P_{X_1^N}(x_1^N) \\ &\quad D_t(x_t, F_{E,t}(x_1^N, m)) \\ &= \frac{1}{N} \sum_{t=1}^N \sum_{m \in \mathcal{M}} \sum_{x_1^N \in \mathcal{X}_1^N} P_M(m) P_{X_1^N}(x_1^N) \\ &\quad D_t(x_t, F_{E,t}(x_1^N, m)) \\ &= \frac{1}{N} \sum_{t=1}^N \mathbb{E} [D_t(X_t, F_{E,t}(X_1^N, M))] = \Delta, \end{aligned} \quad (85)$$

where $Y_1^N = F_E(X_1^N, M)$, $F_{E,t} : \mathcal{X}_1^N \times \mathcal{M} \rightarrow \mathcal{Y}_t$ denotes the t -th symbol generated by the encoding function, and the third line is obtained by the special case of Tonelli's Theorem. If as $N \rightarrow \infty$, the extraction error probability tends to 0, we say that the code achieves the pair $(R = \frac{H(M)}{N}, \Delta)$. Our aim is to show that as $N \rightarrow \infty$, to make sure that the extraction error probability tends to 0, the embedding rate

$$R = \frac{H(M)}{N} \leq R_D(\Delta). \quad (86)$$

First, we write

$$NR = H(M) = I(M; \hat{Y}_1^N) + H(M|\hat{Y}_1^N). \quad (87)$$

Then, following the chain rule of Shannon's mutual information, we can decompose the term $I(M; \hat{Y}_1^N)$ as follows.

$$I(M; \hat{Y}_1^N) = \sum_{t=1}^N I(M; \hat{Y}_t | \hat{Y}_1^{t-1}) \quad (88)$$

$$= \sum_{t=1}^N \left(I(M, \hat{Y}_1^{t-1}; \hat{Y}_t) - I(\hat{Y}_1^{t-1}; \hat{Y}_t) \right) \quad (89)$$

$$\leq \sum_{t=1}^N I(M, \hat{Y}_1^{t-1}; \hat{Y}_t) \quad (90)$$

$$= \sum_{t=1}^N \left(I(M, \hat{Y}_1^{t-1}, X_{t+1}^N; \hat{Y}_t) - I(X_{t+1}^N; \hat{Y}_t | M, \hat{Y}_1^{t-1}) \right) \quad (91)$$

$$= \sum_{t=1}^N \left(I(M, \hat{Y}_1^{t-1}, X_{t+1}^N; \hat{Y}_t) - I(X_t; \hat{Y}_1^{t-1} | M, X_{t+1}^N) \right) \quad (92)$$

$$= \sum_{t=1}^N \left(I(M, \hat{Y}_1^{t-1}, X_{t+1}^N; \hat{Y}_t) - \left(I(X_t; \hat{Y}_1^{t-1}, M, X_{t+1}^N) - I(X_t; M, X_{t+1}^N) \right) \right) \quad (93)$$

$$= \sum_{t=1}^N I(M, \hat{Y}_1^{t-1}, X_{t+1}^N; \hat{Y}_t) - I(X_t; \hat{Y}_1^{t-1}, M, X_{t+1}^N), \quad (94)$$

where (88), (89), (91), (93) are obtained by the chain rule of Shannon's mutual information, (92) is obtained by the Kerner-Martón identity (Csiszár sum identity) [49], and (94) follows from the fact that X_1^N is i.i.d. and M is independent of X_1^N . Here we introduce a sequence of auxiliary RVs $\{Z_t = (M, \hat{Y}_1^{t-1}, X_{t+1}^N)\}_t$. Then, (94) can be written into

$$\begin{aligned} I(M; \hat{Y}_1^N) &\leq \sum_{t=1}^N I(M, \hat{Y}_1^{t-1}, X_{t+1}^N; \hat{Y}_t) - I(X_t; \hat{Y}_1^{t-1}, M, X_{t+1}^N) \\ &= \sum_{t=1}^N \left(I(\hat{Y}_t; Z_t) - I(X_t; Z_t) \right). \end{aligned} \quad (95)$$

By substituting (95) into (87), we have

$$H(M) \leq \sum_{t=1}^N \left(I(\hat{Y}_t; Z_t) - I(X_t; Z_t) \right) + H(M | \hat{Y}_1^N). \quad (96)$$

Following (85), define

$$\Delta_t \triangleq \mathbb{E} [D_t (X_t, F_{E,t}(X_1^N, M))].$$

Then, $\Delta = \frac{1}{N} \sum_{t=1}^N \Delta_t$. By the definition of the rate-distortion bound (7), we have

$$\begin{aligned} \lim_{N \rightarrow \infty} \frac{H(M)}{N} &\leq \lim_{N \rightarrow \infty} \frac{\sum_{t=1}^N \left(I(\hat{Y}_t; Z_t) - I(X_t; Z_t) \right) + H(M | \hat{Y}_1^N)}{N} \\ &\leq \lim_{N \rightarrow \infty} \frac{\sum_{t=1}^N R_{D_t}(\Delta_t) + H(M | \hat{Y}_1^N)}{N} \\ &\leq R_D(\Delta) + \lim_{N \rightarrow \infty} \frac{1}{N} H(M | \hat{Y}_1^N) \\ &\leq R_D(\Delta) + \lim_{N \rightarrow \infty} \Pr(M \neq \hat{M}) \frac{H(M)}{N} + \frac{1}{N}, \end{aligned} \quad (97)$$

where the third line is obtained by (75), the supremum in (7), Corollary 1, and Lemma 4, the fourth line follows from (73), (74) and Lemma 3, and the fifth line is by Fano's inequality. If $R > R_D(\Delta)$, as $N \rightarrow \infty$, (97) would turn into

$$\begin{aligned} \lim_{N \rightarrow \infty} (R - R_D(\Delta)) &\leq \lim_{N \rightarrow \infty} \left(R \Pr(M \neq \hat{M}) + \frac{1}{N} \right) \\ \lim_{N \rightarrow \infty} \Pr(M \neq \hat{M}) &\geq \lim_{N \rightarrow \infty} \frac{R - R_D(\Delta)}{R} = \frac{R - R_D(\Delta)}{R} \\ &> 0. \end{aligned} \quad (98)$$

Therefore, to make sure that as length N tends to infinity, the error probability tends to 0, $R \leq R_D(\Delta)$ must hold, which completes the proof. $\square$

VI. CONCLUSION AND FUTURE DIRECTIONS

In this paper, robust adaptive steganographic coding is investigated. It is formalized as a rate-distortion problem of determining the supremum of all achievable embedding rates of a specialized communication system characterized by a normalized adaptive distortion function and a non-stationary, memoryless sequence of discrete channels, ensuring error-free transmission of the secret message. The optimal theoretical rate-distortion bound is established, accompanied by proofs of its achievability and optimality. By modeling the noisy channels encountered in real-world applications as non-stationary discrete memoryless channels, our derived bound serves as a benchmark for evaluating existing robust methods and provides a target for the

$$\sup_{\{P_{U_t, X_t, Y_t, \hat{Y}_t}\}_{t=1}^N, \mathbb{E}[D(\mathbf{X}, \mathbf{Y})] \leq d} \overline{I(\hat{\mathbf{Y}}; \mathbf{U})} - \overline{I(\mathbf{X}; \mathbf{U})} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \sup_{P_{U_t, X_t, Y_t, \hat{Y}_t}, \mathbb{E}[D_t(X_t, Y_t)] \leq d_t} \left(I(\hat{Y}_t; U_t) - I(X_t; U_t) \right) \quad (82)$$

$$= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \sup_{P_{U_t|X_t}, P_{Y_t|X_t, U_t}, \mathbb{E}[D_t(X_t, Y_t)] \leq d_t} \left(I(\hat{Y}_t; U_t) - I(X_t; U_t) \right) \quad (83)$$

$$= \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \sup_{(P_{U_t|X_t}, f_t) \in \mathcal{P}_{d_t}^{(1)}} \left(I(\hat{Y}_t; U_t) - I(X_t; U_t) \right) \leq R_D(d) = \sup_{(\{P_{U_t|X_t}\}_{t=1}^\infty, \{f_t\}_{t=1}^\infty) \in \mathcal{P}_d} \overline{I(\hat{\mathbf{Y}}; \mathbf{U})} - \overline{I(\mathbf{X}; \mathbf{U})}. \quad (84)$$

development of new practical robust adaptive steganographic coding schemes.

Although the theoretical limit has been established, there are still many problems to be solved in the future. Here we briefly state some of the open problems.

- A major open problem is how to design near-optimal or even provably optimal robust adaptive steganographic codes and coding methods to approach the theoretical limit. The achievability proof of the bound implies that the optimal code can potentially be constructed using a nested code structure. How to design such codes and coding methods adapting to any adaptive distortion function and any non-stationary discrete memoryless channels is an important and challenging area for further research.
- Another problem of interest is how to efficiently calculate the bound for general adaptive distortion functions and non-stationary discrete memoryless channels in practice. For special cases, e.g., the binary Hamming case with the Hamming distortion measure and the stationary BSC(q), the bound can be easily calculated following the traditional approach as our bound agrees with the traditional bound. However, for general adaptive distortion functions and non-stationary discrete memoryless channels, how to solve the optimization problem and thereby explicitly obtain the bound is still an open problem. A solution to this problem will be beneficial for the design of practical robust adaptive steganographic coding algorithms.
- In addition, to make steganography practical, “how to build up the mathematical model of the lossy operations in real-world applications” is also an important problem. For instance, in image steganography, the stego is likely to suffer from JPEG compression as the JPEG format is the most widely-used format of images on the internet. So, whether JPEG compression can be modeled as non-stationary discrete memoryless channels or other probabilistic channels is an important problem and a solution to this will benefit the research field of image steganography both theoretically and practically.
- Following the previous real-world channel modeling problem, in many practical scenarios, the communication channel exhibits memory, a feature that falls outside the non-stationary discrete memoryless channel model adopted in this work. Likewise, natural signals often possess inherent dependencies, making the i.n.i.d. assumption for the cover sequence equally restrictive. Extending the proposed model and the corresponding theoretical bound to accommodate channels with memory and dependent cover sequences is therefore an important direction for future research.

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Qiyi Yao (Member, IEEE) received the BEng degree in computer science and technology from Sun Yat-sen University in 2020 and the PhD degree in cyber science and technology (cyberspace security) from the University of Science and Technology of China in 2025. He is currently a postdoctoral researcher with the Department of Electrical and Electronics Engineering, Bilkent University. His research interests include the intersection of the theoretical and applied domains of computer science, particularly in steganography, coding theory, and DNA-based data storage systems.



Weiming Zhang received the MS and PhD degrees from the Zhengzhou Information Science and Technology Institute, China, in 2002 and 2005, respectively. He is currently a professor with the School of Information Science and Technology, University of Science and Technology of China. His research interests include information hiding coding and multimedia security.



Kai Zeng received the BS and PhD degrees from the University of Science and Technology of China, in 2019 and 2024, respectively. He is currently a postdoctoral researcher with the Department of Information Engineering and Mathematics, University of Siena, Siena, Italy. His research interests include information hiding and multimedia security.



Kejiang Chen (Senior Member, IEEE) received the BE degree from Shanghai University in 2015 and the PhD degree from the University of Science and Technology of China in 2020. He is currently an associate professor with the University of Science and Technology of China. His research interests include information hiding and AI security.